

Beyond binary outcomes: Evaluating and mitigating bias in national standardized test score prediction

Lin Li^a , Namrata Srivastava^{a, b} , Jia Rong^a , Quanlong Guan^c , Dragan Gašević^a ,
Guanliang Chen^{a, *}

^a Centre for Learning Analytics, Monash University, Clayton, VIC, Australia

^b Department of Computer Science, Vanderbilt University, Nashville, Tennessee, USA

^c College of Information Science and Technology, Jinan University, Guangzhou, Tianhe District, China

HIGHLIGHTS

- Measuring bias in continuous educational predictions should go beyond simple group averages to account for differences across the full distribution between groups.
- We propose AreaCDF and AreaPDF to capture predictive disparities between two groups over continuous outcomes across the full distribution.
- In continuous educational predictions, achieving balanced demographic group representation does not guarantee fair predictive performance across groups.
- Fairness interventions should be tailored to data and models, considering whether demographic imbalances are global, local, or intertwined with skewed score distributions.

ARTICLE INFO

Keywords:

Ethical learning analytics
Early childhood education
Predictive bias
Debiasing

ABSTRACT

Predictive analytics in education holds promise for supporting student learning but poses risks of bias potentially disadvantaging certain groups based on demographic attributes such as gender. Prior work on fairness in educational prediction has largely focused on binary outcomes in higher education, leaving continuous predictions, such as standardized test scores, comparatively understudied, especially in early schooling where consequences can be long-lasting. Despite a few fairness metrics being available for continuous outcomes, they often rely on mean differences, obscuring distributional disparities that may disadvantage specific students. Even fewer studies have attempted to address bias in continuous predictions. This study addresses these gaps by introducing a fairness evaluation framework grounded in statistical distance measures with two adapted metrics, AreaPDF and AreaCDF, that capture group disparities across the full distribution of continuous predicted outcomes. We evaluate these metrics in the context of predicting national standardized test scores in early childhood education and further develop five reweighting-based data-balancing methods to mitigate bias in continuous prediction tasks. Empirical analyses show that the proposed metrics reveal biases overlooked by traditional measures and that the reweighting strategies substantially improve fairness, underscoring the importance of aligning debiasing methods with demographic and score-distribution characteristics.

1. Introduction

Predictive bias or unfairness in education commonly refers to systematic disparities in predictions made by data-driven models that may disproportionately affect students based on their sensitive attributes such as race, gender, or socioeconomic status (Baker & Hawn, 2021; Chinta et al., 2024). Most existing studies on this topic have focused

on binary predictive tasks, such as predicting whether a student will drop out of a course (Lallé et al., 2024) or re-enroll in a school (Yao et al., 2025). Predictive bias in such contexts has been measured from various perspectives, with one of the most concerning biases being the significantly different prediction error distributions across demographic groups (Cohausz et al., 2024; Gardner et al., 2019). When predictions exhibit such bias, i.e., disparities in error distributions across

* Corresponding author.

Email addresses: lin.li@monash.edu (L. Li), namrata.srivastava@vanderbilt.edu (N. Srivastava), jiarong@acm.org (J. Rong), gql@jnu.edu.cn (Q. Guan), dragan.gasevic@monash.edu (D. Gašević), guanliang.chen@monash.edu (G. Chen).

<https://doi.org/10.1016/j.caeai.2026.100585>

Received 17 June 2025; Received in revised form 21 March 2026; Accepted 23 March 2026

Available online 27 March 2026

2666-920X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

groups, are used to allocate resources to students, certain groups may be systematically misaligned with the support they require (Barocas et al., 2023). This misalignment risks creating or exacerbating gaps, undermining the core purpose of education: ensuring equal opportunities for all students to succeed (Culp & Drerup, 2024; Kizilcec & Lee, 2022). In response to these concerns, researchers have dedicated significant efforts to identifying and mitigating such biases, particularly in binary decision-making scenarios (Deho et al., 2022; Li et al., 2025).

However, risks associated with such predictive bias extend beyond binary outcomes in education. Unlike binary outcomes, continuous scores that are capable of capturing each student's specific situation, are more prevalent in real-world educational scenarios, and are often preferred for tailoring personalized learning support (Glaser et al., 2001; Kuncel & Hezlett, 2007). A prominent example is standardized testing (Friedman et al., 2025), where early predictions of student test scores may guide educators in designing differentiated instruction or targeted interventions to better meet students' diverse needs (Li et al., 2024; Ritter et al., 2013). In such contexts, prediction errors, measured as numerical deviations between predicted scores and actual scores, serve as indicators of how accurately a student's true competency is assessed. Just as bias in binary predictions arises when error rates (e.g., false positives) differ across demographic groups, similar biases can emerge in continuous score predictions when different groups experience unequal likelihoods of receiving the same level of prediction error. For instance, in the case of early predictions for standardized test scores, if 20% of female students experience a prediction error of at least 10 marks, compared to just 5% of male students, female students are more likely to be disproportionately misaligned with the support that best matches their needs, potentially widening gender disparities in learning outcomes. Therefore, addressing bias in continuous predictions is particularly important to ensure fair and effective educational interventions.

Nevertheless, addressing predictive bias in continuous scores presents significant challenges. The first major challenge lies in measuring such bias. Unlike binary prediction scenarios, where errors correspond to discrete misclassifications (e.g., false positives), prediction errors in continuous scores are numerical values that span a continuous range. As a result, while disparities in binary prediction errors can be clearly mapped to differences in the rates of a few specific error types, disparities in continuous predictions may emerge across various points in the entire error distribution, making them more complex to capture and quantify. Existing metrics attempt to capture these disparities but often fall short of fully reflecting the underlying distributional differences between groups. For example, metrics such as mean error difference directly compare central tendencies (Gursoy & Kakadiaris, 2022; Mohamed & Schuller, 2022) of errors between groups, overlooking the variability of errors within each demographic group. Other metrics such as R^2 -based ones proposed for automated essay scoring tasks (Lee et al., 2025; Litman et al., 2021), treat bias as the proportion of variance in prediction errors that can be explained by group membership, similarly failing to capture finer-grained disparities regarding how errors are distributed differently across groups. Consequently, there remains a significant gap in developing more comprehensive metrics for fully assessing distributional disparities in continuous predictions. Additionally, research on mitigating biases in continuous predictions remains limited, particularly regarding the effectiveness of training data processing (Li et al., 2023). While training data imbalance is widely acknowledged as a primary source of predictive bias in binary tasks (Liu et al., 2025b; Sha et al., 2022), its role in shaping bias in continuous predictions has yet to be systematically explored.

Toward this end, by focusing on one representative yet under-investigated task in early childhood education, i.e., predicting students' performance on numeracy and literacy (reading) in a national standardized test, this study tackled the above challenges to address predictive bias in continuous predictions with the following contributions. First,

motivated by the widespread use of statistical distance measures for comparing distributions in machine learning (Basu et al., 2011), this study proposes a novel fairness evaluation framework for continuous prediction by adapting two well-established statistical distance metrics to the educational context, namely *Wasserstein Distance* (Panaretos & Zemel, 2019) and *Total Variation Distance* (Rassouli & Gündüz, 2019). We operationalize these distances for group fairness analysis through their area-based interpretation with the new framings, i.e., Area Between Cumulative Distribution Functions (AreaCDF) and the Area Between Probability Density Functions (AreaPDF), to highlight their natural suitability for full distributional comparisons.¹ We also provide theoretical analysis showing how these metrics reveal systematic distributional bias that may be overlooked by conventional aggregated fairness measures. Building on these two metrics, the framework further integrates a multi-dimensional bias evaluation that assesses predictive bias across four complementary dimensions, including overall score accuracy, overall score difference, conditional score accuracy and conditional score difference. Together, these components form a single, coherent framework for diagnosing fairness in continuous prediction. Using this framework, we conducted an empirical analysis of bias related to students' gender and disability status when predicting their numeracy and reading test scores, demonstrating that it reveals fairness patterns that remain hidden under conventional mean-based metrics. Finally, based on the assumption that the predictive bias might be largely attributed to the under-representation or over-representation of certain score-group combinations in the training data, this study further proposes five debiasing approaches that balance training samples based on both group membership and score distribution through reweighting. These approaches improve predictive fairness without imposing a strict accuracy trade-off, demonstrating their practical value for supporting more equitable and reliable predictive systems in educational settings.

To summarize, this study aimed to answer two research questions: **RQ1:** To what extent can bias in continuous predictions, specifically in the context of student test-score prediction, be uncovered using the proposed fairness evaluation framework, and how do these findings compare with traditional mean-based fairness assessments? **RQ2:** How effectively do the proposed pre-processing debiasing methods reduce predictive bias, and what are the resulting trade-offs regarding predictive accuracy?

2. Related work

2.1. Measuring bias in continuous educational predictions

Fairness of machine-generated predictions is often discussed at either the individual or group level (Caton & Haas, 2020), where individual fairness requires similar predictions for similar individuals and group fairness emphasizes the overall similarity of predictions across different demographic groups. This study specifically focuses on group fairness. In a broader sense, group fairness has been commonly interpreted as the requirement that the "quality" of predictions should be similar across different groups defined by sensitive attributes like gender or race. The "quality" here can correspond to a wide range of predictive performance-related metrics that indicate overall alignment or misalignment between predictions and actual outcomes, such as true positive rates or false positive rates (Caton & Haas, 2020; Hardt et al., 2016). In the field of education, most work is centered around binary classification tasks such as identifying students at risk of dropping out of a school or course,

¹ Note that we do not introduce AreaPDF or AreaCDF as new statistical distances; rather, we adapt their area-based formulations to highlight their natural suitability for fairness evaluation in continuous prediction settings. This framing captures disparities across the full score distribution, addressing limitations of conventional aggregated metrics such as R^2 -based ones for continuous outcomes.

with a major focus on developing metrics examining bias in binary predictions (Verger et al., 2023). Notable advancements in this line of work include ABROCA (Gardner et al., 2019) and MADD (Verger et al., 2023), which extend fairness assessment from comparing single metrics of predictive performance (e.g., Equal Opportunity (Hardt et al., 2016)) to comparing the full distributions of prediction-related scores across demographic groups. These methods move beyond simple aggregated comparisons, offering more nuanced insights into the disparities between groups by evaluating the entire distribution of predicted outcomes.

Despite these advancements, very few of them are applicable to bias detection for non-binary predictive tasks, especially for continuous predictions (Li et al., 2023). This gap is particularly significant given the prevalence and importance of continuous predictions in guiding the provision of finer-grained interventions and personalized support in education (Kuncel & Hezlett, 2007). At a basic level, some studies directly compare *Mean Difference* in predictive performance such as Mean Absolute Error (MAE) across groups (Dikli & Bleyle, 2014). On a more granular level, some studies (Loukina et al., 2019; Yang et al., 2024) assess the predictive power of group membership for predicting predicted outcomes, using R^2 as a measurement of bias. This approach was initially applied to examine biases in automated essay scores across different dimensions, such as whether score accuracy (e.g., squared prediction error) or score difference (e.g., signed prediction error) differs across different demographic groups. Bias metrics for these dimensions are all calculated as R^2 of fitting the specific predicted outcome using demographic group membership (detailed in Appendix B.1). These metrics, despite being originally designed for ordinal scores, can also be easily applied to continuous scores. However, a significant limitation of these metrics is that they essentially reduce to a combination of group-averaged outcomes when demographic group membership is binary, as will be mathematically proven later in this study (Appendix B.2). For continuous scores, relying on group-averaged outcomes can be problematic, as it may obscure important variations between groups, as will be discussed in Section 4. Moreover, such metrics can be challenging for educators to interpret and act upon, as they only provide a single value, making it difficult for stakeholders to understand the specifics of how bias manifests in predictions. Therefore, it is crucial to develop metrics that capture differences in predicted outcomes between groups at a finer-grained level, offering educators more actionable insights and greater flexibility in tailoring interventions.

2.2. Comparing distributions via statistical distance measures

Statistical distance measures broadly refer to metrics that quantify how one probability distribution differs from another (Basu et al., 2011; Cai & Lim, 2022). Given their ability to compare entire distributions instead of only summary statistics like means or variances, they have served as the foundation for many machine-learning techniques (Liu et al., 2025a; Shen et al., 2018). Well-established examples include *Wasserstein distance* and *Total Variation (TV) distance*, to name just a few. Among these, *Wasserstein distance* and *Total Variation distance* are primary representatives of two main paradigms for comparing probability distributions. *Total Variation distance* (Rassouli & Gündüz, 2019) views distributions through a “vertical” paradigm, comparing the probability density values at each point in the sample space. Mathematically, it is defined as the supremum difference between the probabilities assigned to any measurable event, which for continuous variables is equivalent to half of the area between two density functions, i.e., the integral over the vertical distance between two probability density functions for all points. The distance reaches its maximal value of 1 for any two distributions with disjoint supports, regardless of how far apart the distributions are, indicating that TV is insensitive to horizontal displacement. *Wasserstein distance* (Shen et al., 2018), in a different vein,

incorporates horizontal geometry by treating distribution shifts as an optimal transport problem. It defines distance as the minimum effort required to reconfigure one distribution into another, accounting for both the amount of probability mass moved and the distance over which it is transported. In one-dimensional cases, this distance geometrically represents the area between two cumulative distribution functions (CDFs). In machine learning, TV distance has been used to quantify the differential privacy of algorithms when evaluated on neighboring datasets (Rassouli & Gündüz, 2019), whereas Wasserstein distance has been widely applied in areas such as generative modeling and domain adaptation to guide models toward outputs that more closely align with real data or a target domain (Deshpande et al., 2018; Shen et al., 2018). This complementary perspective on distributional disparity aligns closely with our goal of assessing fairness between two groups’ continuous predictions over full distributions. In addition, the visual mapping of these metrics to the areas between the corresponding PDFs and CDFs provides an intuitive way to reveal where bias occurs, making the results easier to communicate and interpret for researchers, educators, and practitioners. These advantages motivate our investigation into adapting these two metrics for measuring fairness in continuous predictions within educational contexts.

2.3. Mitigating predictive bias in educational predictions

Predictive bias is often assumed to originate from either the training data or the model itself, with training data widely acknowledged as the predominant source of bias (Caton & Haas, 2020). In particular, training data distributional imbalances, such as the disproportionate representation of demographic groups or the unequal association of task labels (e.g., class labels for classification or task scores for regression) with certain groups, have been shown to induce predictive bias. A simple strategy for addressing demographic imbalance is to enforce equal representation across demographic groups by up-weighting minority samples and down-weighting majority samples, under the assumption that bias arises from unequal group frequencies. When both demographic labels and class labels are considered, one straightforward extension is to enforce equal representation across all groups defined by the combination of class labels and demographic memberships (Lahoti et al., 2020). This approach can be viewed as a global, or absolute, joint balance, because it forces the overall distribution of samples across all label-group combinations to be uniform, regardless of how those combinations occur naturally in actual data. To relax this strong assumption, which often does not hold in practice, ideas have also been proposed to restrict joint balance to within subgroups, such as requiring equal representation of class labels within each demographic group, or vice versa (Han et al., 2022). A closely related but conceptually distinct line of work is the reweighting method proposed in Kamiran and Calders (2012), which assumes independence between class distribution and demographic group membership for bias-free data. Resampling- and reweighting-based methods that operationalize these ideas have been developed and demonstrated to be effective across various educational tasks (Li et al., 2023; Sha et al., 2022).

However, these ideas have so far been operationalized primarily for mitigating bias in binary prediction settings. For continuous prediction tasks, most fairness-improving studies focus on model-level modifications, either by constraining the loss function or altering the optimization process to balance training across different demographic subgroups. For example, *FairReg* (Berk et al., 2017) introduces a regularization term to the objective function of a linear regression model, penalizing it for treating similarly scored instances from different demographic groups differently. *Reduction* (Agarwal et al., 2019) is another well-established debiasing approach that enforces fairness by constraining the optimization process to bound group-wise loss. Despite their effectiveness, these approaches are inherently model-dependent and are not readily transferable to other model classes (e.g., tree-based or

non-parametric models), limiting their general applicability. In contrast, data-level debiasing provides a more flexible and model-agnostic approach, yet remains relatively underexplored in continuous prediction settings. Only one recent study outside the education domain (Mohamed & Schuller, 2022) can be found with two data-balancing strategies evaluated, namely DemoRW and ReLabel. The former balances demographic representation globally through reweighting, whereas the latter adjusts observed scores so that different groups share the same mean and standard deviation. Despite these initial efforts, no prior work has systematically examined how training-data imbalances influence bias in continuous predictions, nor how such bias can be effectively mitigated. As a result, it remains unclear whether data-balancing approaches developed for classification tasks can generalize to continuous prediction problems. This gap highlights the need for bias-mitigation methods tailored specifically to continuous predictions in education, so that predictive models provide more equitable outcomes across diverse student populations. Toward this goal, this study was also inspired to adapt existing data-balancing ideas for continuous outcomes in order to examine their effectiveness in reducing predictive bias.

3. Task, dataset and model

Task. This paper focuses on a national standardized test that assesses students' numeracy and reading competency, which is mandatory for students in the early school years, such as Years 3, 5, or 7. National standardized tests have long been administered in many countries at various stages before higher education, playing a crucial role in tracking student learning progress, identifying achievement gaps, and informing educational policies (Wiliam, 2010). Early prediction of students' scores in these assessments is particularly valuable for both students and schools, as it may enable timely interventions, targeted support, and data-driven decision-making to enhance learning outcomes (Li et al., 2024). Hence, this study aims to predict students' scores in numeracy and reading tests in Year 3 based on their performance in school-administered formative assessments prior to Year 3.

Dataset. The dataset used in this study was collected from a large school system covering 58 primary schools in Australia, where students' test scores in Year 3 and performance across 12 different formative classroom assessments in Kindergarten, Year 1 and Year 2 were recorded. Data about students' demographics including gender and disability, their family background, and school-level information (e.g., previous overall test performance) were also available. Statistics about demographic representation and scores are provided in Table 1. More details can be found in Appendix A.

Predictive Model. The primary objective of this study is to measure and improve fairness in continuous predictions, specifically the predicted student scores in national numeracy and reading tests, by (a) first identifying biases across different task-attribute scenarios, where the "task" refers to the prediction target (numeracy or reading scores), and the "attribute" refers to the demographic attribute under examination for bias (gender or disability), and (b) then, addressing training data distributional imbalances to mitigate the biases. Rather than optimizing accuracy via extensive model comparisons, we prioritize fairness and evaluate three representative prediction models, i.e., linear regression

(LR) (Montgomery et al., 2021), multilayer perceptron (MLP) (Murtagh, 1991), and XGBoost (Chen & Guestrin, 2016), to assess the robustness of fairness metrics and debiasing strategies across different classes of models. LR provides a transparent, commonly used baseline for continuous educational predictions and fairness evaluation (Berk et al., 2017; Oliveira et al., 2023), while MLP and XGBoost introduce nonlinear capacity and interaction modeling, enabling us to examine whether fairness findings and mitigation effects persist beyond linear assumptions. To build the predictive model, we engineered three groups of features: (1) *Formative Assessment Performance*, (2) *Family Background*, and (3) *School Characteristics*. The *Formative Assessment Performance* features consist of 52 numerical variables representing student scores on 12 formative assessments administered during Kindergarten, Year 1, and Year 2. Further details on feature engineering can be found in Appendix A.

4. Measuring predictive bias through full distributional disparities

This section first outlines the motivation and key considerations underlying metrics that capture full distributional disparities (Section 4.1), then introduces the formalization of the proposed metrics (Section 4.2), and finally describes their advantages over existing metrics (Section 4.3).

4.1. Motivation and key considerations

Before introducing the proposed metrics, this sub-section first outlines the motivation behind their development, then discusses key considerations that inform their design and use for bias evaluation.

4.1.1. Why full distributional disparities matters?

Traditional metrics often evaluate bias as the difference between two single values that summarize the overall predictive performance for each group, such as group difference in mean absolute error. However, this "averaging effect" can be problematic, as it risks masking underlying biases where the two groups actually differ significantly in their likelihoods of receiving predictions of similar quality. For instance, two groups might exhibit identical MAE values, yet one group could experience small and consistent errors, while the other suffers from a mix of extreme over- and under-predictions. A simple example is illustrated in Table 2, where both groups have an identical MAE of 4, but their absolute error distributions differ significantly. Group A's predictions are relatively closer to the true values, with no prediction exceeding an absolute error of 10. In contrast, 40% of Group B's predictions deviate by 100 marks from the actual scores. In education, this discrepancy is not trivial, as it suggests that one group of students is more likely to receive misaligned learning support than the other. While metrics based on R^2 , such as R^2 -OSA, R^2 -OSD, and R^2 -CSD (detailed in Appendix B.1), do not directly rely on group mean differences and instead quantify bias as the extent to which demographic group membership explains variance in predicted outcomes, our **mathematical proof** shows that when the demographic attribute is binary, these metrics reduce to a combination

Table 2

An example illustrating the drawback of measuring bias as the difference in group-averaged outcomes such as MAE. Despite both groups (Group A and Group B) having identical MAE values and the same sample size (each consisting of 1000 samples), their error distributions differ significantly. Group A has more errors concentrated around smaller values, while Group B has a larger proportion of extreme errors, highlighting how group-averaged metrics may overlook important disparities in error distributions.

Table 1

Basic statistics about demographic representation and score means of the training data.

Statistics	Gender		Disability	
	Male	Female	Without disabilities	With disabilities
#Samples	919	889	1247	516
Mean Numeracy	0.390	0.383	0.399	0.358
Mean Reading	0.384	0.394	0.404	0.357

of group-averaged outcomes, see a brief summary of the proof below and a complete proof in [Appendix B.2](#).

Theorem 1. Consider a simple linear regression model with the form $\hat{y} = w_0 + w_1 x$, where $\hat{y}$ is the predicted score outcome such as prediction errors, x is a binary independent variable indicating demographic group membership, (e.g., $x \in \{0, 1\}$ for male vs. female), w_0 is the intercept, and w_1 is the regression coefficient. Let y be the true outcome values. Then, the coefficient of determination R^2 of this model is a function of the difference between the group-wise average values of the actual scores y :

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_i y_i^2 - [n_1(\bar{y}_1^2 - \bar{y}_0^2) + n \cdot \bar{y}_0^2]}{\sum_i (y_i - \bar{y})^2}. \quad (1)$$

Based on the above, for overall score difference (OSD) where y represents the actual prediction error, R^2 -OSD essentially measures the difference between the squared average group prediction errors. However, it does not capture distributional disparities beyond the group means, and therefore may overlook fairness differences that exist within or across the full range of prediction outcomes. A similar conclusion applies to R^2 -OSA and R^2 -CSD.

Therefore, these R^2 -based metrics carry similar risks of obscuring the full extent of distributional disparities between groups.

4.1.2. Unpacking full distributional disparities

Examining full distributional disparities in continuous predictive outcomes between groups naturally raises two key considerations: (1) which dimensions of the predicted outcomes should be considered for bias detection and (2) how the full distribution of predicted outcomes should be represented to effectively highlight disparities? The first consideration emphasizes the need to define which predicted outcomes should be examined and how groups of these values can be analyzed. This need arises because: (1) for continuous predictions, whether signed or absolute, different meanings are conveyed, where signed errors indicate whether predictions over- or under-estimate the true values (i.e., score difference), while absolute errors focus solely on the accuracy (i.e., score accuracy), disregarding the direction of the error, and (2) these values, when compared by conditioning on students with similar actual scores versus not, also convey different insights. Without conditioning, they reveal overall group disparities in score accuracy (using absolute error) or score difference (using signed error), that is, whether accuracy or deviation in predictions is similarly distributed across demographic groups. When conditioning on actual scores, the comparison highlights whether disparities remain among students who perform at

similar levels but are from different demographic groups, offering a more detailed view of potential bias. Hence, by considering (1) whether the focus is on score accuracy (absolute errors) or score difference (signed errors), and (2) whether comparisons are made across all students (overall) or conditioned on students with similar actual scores (conditional), predictive bias can be evaluated across four dimensions: overall score accuracy, overall score difference, conditional score accuracy, and conditional score difference. Previous studies have examined bias across three of them using R^2 as introduced in [Appendix B](#). Once the dimension of predictive outcomes to examine is determined, the second consideration arises from the fact that the distribution of a continuous variable can be represented in two main forms: the cumulative distribution function (CDF) and the probability density function (PDF) (see [Fig. 1](#)). CDF gives a cumulative view of the variable, showing the probability of being less than or equal to specific values. This is particularly useful for educators concerned with overall trends at defined thresholds. For example, CDF can be used to assess whether a larger proportion of female students are under-predicted by more than 100 marks compared to their male counterparts. A significant bias is alerted if the proportion of female students below this threshold exceeds that of male students by a notable margin (determined by domain expertise). Conversely, PDF provides a detailed view of how examined outcomes are distributed across the whole range, enabling precise identification of ranges where disparities are most pronounced. For example, it can reveal whether male students are disproportionately more likely to receive near-zero prediction errors compared to female students.

Therefore, building on these foundations, this study was conducted to examine bias across four different dimensions of predicted outcomes, including Overall Score Accuracy, Overall Score Difference, Conditional Score Accuracy, and Conditional Score Difference. For each of these dimensions, the distribution of the examined outcome variable (i.e., score accuracy or score difference) can be represented in the form of a cumulative distribution function (CDF) or probability density function (PDF).

4.2. Proposed metrics

From a geometrical perspective, these disparities can be interpreted as the area between the curves representing the two distributions. Therefore, motivated by the idea of metrics like *ABROCA* ([Gardner et al., 2019](#)), which captures bias as the area between two curves describing error distributions, this study introduces *AreaCDF* and *AreaPDF* to measure differences in outcome variables between two groups by calculating

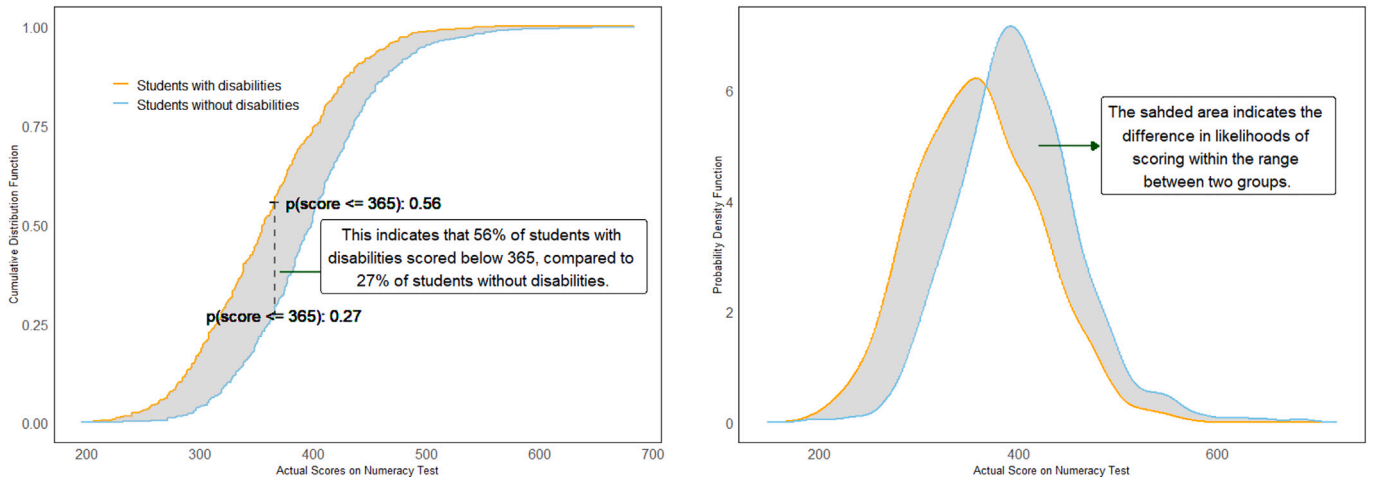


Fig. 1. The left figure shows the cumulative distribution functions (CDF) of numeracy scores for students with and without disabilities, with the shaded region indicating the difference between the two CDFs (i.e., *AreaCDF*). The right figure shows the probability density functions (PDF) for the same set of scores, with the shaded region indicating the difference between the two PDFs (i.e., *AreaPDF*).

the area between their cumulative distribution functions (CDFs) and probability density functions (PDFs), respectively. Denote the outcome variable to be evaluated as x (e.g., score accuracy or score difference), *AreaCDF* is calculated by integrating the absolute difference between the cumulative distribution functions of two groups across all possible values:

$$\text{AreaCDF}(x_{G_b}, x_{G_c}) = \int_{-\infty}^{+\infty} |F_{G_b}(x) - F_{G_c}(x)| dx, \quad (2)$$

where G_b and G_c represent the baseline group and the comparison group, respectively, and $F_{G_b}(x)$ and $F_{G_c}(x)$ are the cumulative distribution functions of the predicted outcome x for group G_b and group G_c , respectively. Noting that *AreaCDF* is equivalent to *first-order Wasserstein distance* (Panaretos & Zemel, 2019), which is a well-established statistical metrics for measuring distance between two probability distributions. A larger *AreaCDF* indicates that the two distributions are more different, implying a higher bias. Theoretically, the metric can increase without bound as the divergence between the distributions grows. However, in practice, this metric is typically constrained by the specific score scale. For instance, if x is standardized test score, *AreaCDF* will not exceed the maximum possible score range. Consequently, the actual range in practice depends on the context and the specific outcomes being measured.

Over the same outcome variable x , *AreaPDF* is calculated by integrating the absolute differences between the probability density curves of the two groups, i.e.,

$$\text{AreaPDF}(x_{G_b}, x_{G_c}) = \int_{-\infty}^{+\infty} |f_{G_b}(x) - f_{G_c}(x)| dx, \quad (3)$$

where $f_{G_b}(x)$ and $f_{G_c}(x)$ represent the probability density functions (PDFs) of score x for the baseline group G_b and the comparison group G_c , respectively. Notably, *AreaPDF* is equivalent to twice the *Total Variation distance* (Gibbs & Su, 2002), which is also a well-established statistical metric for measuring the distance between two probability distributions. *Total Variation distance* ranges from 0 to 1, so *AreaPDF* ranges from 0 to 2. The maximum value, i.e., 2, occurs when the two distributions are completely non-overlapping. For both *AreaCDF* and *AreaPDF*, while the theoretical integration bounds are $-\infty$ and $+\infty$, practical computation is limited to the observed range of x , ensuring relevance to real-world applications.

These two metrics can be applied to any two continuous distributions to quantify their disparities. For example, when applied to actual test scores, as shown in Fig. 1, *AreaCDF* measures group differences in the proportion of students scoring below each possible score value, while *AreaPDF* examines group differences in the proportion of students scoring around each specific value. As introduced in Section 4.1.2, bias will be examined across four dimensions: *Overall Score Accuracy* (OSA), *Overall Score Difference* (OSD), *Conditional Score Accuracy* (CSA), and *Conditional Score Difference* (CSD), for which six metrics are further derived using *AreaCDF* and *AreaPDF* as detailed below.

- For *overall* bias assessments, i.e., OSA and OSD, as discussed in Section 4.1.2, the focus is on comparing predictive performance across all students from different demographic groups. For OSA, *AreaCDF*-OSA and *AreaPDF*-OSA are derived by applying *AreaCDF* and *AreaPDF* to score accuracy (i.e., x in Eqs. 1 and 2 is the absolute prediction error $|\hat{y} - y|$, where $\hat{y}$ is the predicted score and y is the actual score). *AreaCDF*-OSA examines whether the proportions of students with predictions below any given accuracy value are equal across demographic groups, while *AreaPDF*-OSA focuses on the proportions of students with predictions within that same accuracy value. Similarly for OSD, *AreaCDF*-OSD and *AreaPDF*-OSD are derived by focusing on score difference, i.e., signed prediction errors $\hat{y} - y$.
- For *conditional* bias assessments (CSA and CSD), the focus is on comparing predictive performance between students from different demographic groups but with similar actual performance. This

adds an additional condition, focusing on specific subsets of students with comparable actual scores. According to the definition of Eqs. 1 and 2, CDF in this case translates to the form of $F(x_1|x_2)$ and PDF to $f(x_1|x_2)$, where x_1 represents the outcome variable such as score accuracy or score difference, and x_2 represents the actual scores. Calculating *AreaCDF* or *AreaPDF* over such two-dimensional distributions is prohibitively costly and challenging due to the need for joint distribution estimation and the complexities associated with joint integration, which require significant computational resources and sophisticated methods not yet fully matured or optimized for practical use. Hence, this study proposes a simple yet flexible approach to derive *AreaPDF*-CSA and *AreaPDF*-CSD. Specifically, students are first divided into different performance subgroups based on their actual test performance. Then, *AreaPDF* is calculated over the two groups' univariate distributions of the outcome variable (i.e., score accuracy or score difference) within each performance group. The resulting values (i.e., *AreaPDF*-CSA and *AreaPDF*-CSD of each performance subgroup) are aggregated to obtain an overall measure of predictive bias.

Note that for CSD and CSA, *AreaCDF* is not applied. This consideration is because when students are grouped based on their actual scores (i.e., when CSD or CSA is used), some groups may contain very few students, e.g., those with very low or very high actual scores. In such cases, comparisons between CDFs are less meaningful, as CDFs might be unstable and provide unreliable estimates when sample sizes are very small (Gibbons & Chakraborti, 2014). For student grouping in *AreaPDF*-CSA and *AreaPDF*-CSD, the grouping strategy can be flexibly determined based on the needs of real-world educational contexts. For example, when educators are primarily concerned with fairness among students who fall within the same score ranges, groups can be formed using equal-width binning, i.e., dividing students into bins with the same score width. In contrast, when the goal is to examine fairness among students with similar relative standing (such as those in the same rank or percentile bands), grouping can instead be performed using *Equal-Frequency* binning, in which bins are defined according to students' ranks rather than specific scores. In this study, we adopt a 10-level Equal-Width grouping (following the common practice of the 10-band scale used in national testing in Australia (McGaw et al., 2020)), given that the NAPLAN assessment employs fixed score bands. In addition, we also report results using both binning methods, i.e., Equal-Width and Equal-Frequency binning, to better understand how different grouping choices may influence fairness conclusions. It is worth noting that under Equal-Width binning, different subgroups may contain different numbers of students. Although subgroup-size weighting is feasible, we treat all subgroups as equally important and therefore report unweighted averages. This choice reflects the principle that the effects of bias on smaller student groups should not be discounted simply because those groups are smaller in size.

4.3. Advantages over existing fairness metrics for continuous predictions

The nature of distribution-aware measurement of *AreaCDF* and *AreaPDF* offers several advantages relevant to education. (1) As illustrated in Table 2 and the mathematical breakdown proof in Appendix B, metrics directly comparing group-averaged outcomes and widely used R^2 -based metrics all ignore the distributional variations of the outcome values. This limitation is especially concerning in educational settings, where students with top performance or those struggling at the lowest levels often have under-representation in the training data. When such students are concentrated in one demographic group, it is highly likely that this group will have a higher proportion of students with larger prediction errors. In such cases, metrics that compare group-averaged outcomes might report zero bias, even though the error distribution in the other group may differ significantly (e.g., centered on middle-range errors), as long as both groups have the same mean errors. The

distribution-aware measurement adopted by the proposed metrics overcomes this limitation by accounting for differences at each possible outcome value across the full range. (2) Both *AreaCDF* and *AreaPDF* share the characteristic of “threshold-independent” and are rather flexible to adapt to calculate the area between certain score ranges, similar to the ABROCA (Gardner et al., 2019). In real-world applications, educators may prioritize different score thresholds depending on the context. The threshold-independent nature of these metrics enables educators to focus on the range of values that are most relevant to their educational goals. (3) Similar to the “complete” nature of ABROCA, both metrics can be directly computed from the predictive results, without additional data collection. Both metrics simply integrate the absolute difference between CDFs or PDFs, with well-established open-source implementations available for their calculation. (4) Both *AreaCDF* and *AreaPDF* have a straightforward geometric interpretation, which can be easily visualized. This makes it intuitive for educators and researchers to understand the fairness of the predictive model. Additionally, both metrics are linked to mathematically well-defined statistical distance metrics, allowing for more robust statistical analysis, including significance testing.

5. Tackling data imbalance for enhancing predictive fairness

Distributional imbalance in training data has been widely acknowledged as a primary factor for predictive bias, with demographic imbalance and joint imbalance concerning both demographic groups and task labels documented as two main types (Dominguez-Catena et al., 2024). Demographic imbalance refers to situations where certain demographic groups are overrepresented in the training data, such as more males than females. Joint imbalance arises when certain score outcomes or class labels are more frequently associated with particular demographic groups, such as male students being more frequently represented in STEM fields. While addressing these imbalances has proven effective in improving fairness for binary prediction tasks, their effects on continuous predictions remain unknown. The general approach to addressing such imbalances is to ensure a more balanced representation of both over-represented and under-represented samples. One classical method is to reweight the training data by assigning higher weights to underrepresented samples, giving them greater influence in the model training process. Reweighting methods have been proposed to tackle demographic imbalance for continuous predictions (Mohamed & Schuller, 2022), although not specifically in educational contexts. However, joint imbalances concerning both demographic group membership and task labels, while often considered more important, remain unexplored for continuous predictions. Therefore, motivated by previous efforts in balancing training data through reweighting, this study proposes five reweighting approaches that address imbalances in both demographic group membership and task score distribution, each accounting for different distributional characteristics that may contribute to predictive bias.

- **Global Joint Balance (JointBal)**, reweights each sample based on the joint probability of its task score and group membership globally, i.e., $w_{yg} = \frac{1}{p(y,g)}$, where $p(y,g)$ is the probability of a sample having a task score y and group membership g . This approach assigns higher importance to samples with under-represented combinations of task scores and group memberships, focusing on those that are less frequent across the entire dataset. The goal is to balance the representation of all task score and group combinations, ensuring that under-represented samples receive adequate attention during training.
- **Task-aware Joint Balance (Task-JointBal)** extends the concept of JointBal by incorporating the specific distribution of task scores into the reweighting process. It adjusts the sample weights based on the ratio between the marginal probability of a task score $p(y)$ and the joint probability of a task score and group membership $p(y,g)$, i.e., $\frac{p(y)}{p(y,g)}$. Unlike Global Joint Balance, which focuses on

the overall representation of task score and group combinations, Task-JointBal accounts for the frequency of task scores, ensuring that samples with task scores which are globally frequent but underrepresented for specific groups are upweighted.

- **Demographic-aware Joint Balance (Demo-JointBal)** extends on Global Joint Balance but considering the effects of demographic group representation, which reweights each training sample based on the ratio of joint probability of its task score and group membership to the marginal probability of the group membership, i.e., $w_{yg} = \frac{p(g)}{p(y,g)}$. Demo-JointBal accounts for the representation of demographic groups, ensuring that globally under-represented (task score, group) pairs, particularly those from under-represented demographic groups, are re-weighted according to their local frequency within the group.
- **Conditional Demographic Balance (CondDemoBal)**, different from above methods focused on joint probability, reweights each sample based on the ratio of the marginal probability of its task score $p(y)$ to the conditional probability of being from a specific group given the task score $p(g|y)$, i.e., $\frac{p(y)}{p(g|y)}$. This method assumes that, given a task score y , the probability of the sample belonging to any specific group g should be the same for all groups, and equal to the the marginal probability of the task score divided by the number of groups, i.e., $p(g_1|y) = p(g_2|y) = \dots = p(g_n|y) = \frac{p(y)}{n}$. It ensures that samples from under-represented groups within a specific task score are assigned higher weights that are proportional to the overall representation of the task score.
- **Demographic-Task Independence Balance (DTIndBal)**, motivated by *Independence* concept proposed in (Kamiran et al., 2010), assumes that the probability of observing a specific task score should be independent of group membership. Under this assumption, the expected joint probability of observing a (task score, group) pair is the product of the marginal probabilities of the task score and the group, i.e., $p^{exp}(y,g) = p(y) \cdot p(g)$. Hence this method reweights each sample based on the ratio of expected probability to the observed probability, i.e., $w_{yg} = \frac{p^{exp}(y,g)}{p^{obs}(y,g)}$.

The five reweighting strategies differ primarily in the balancing target, that is, the aspect of the data they aim to equalize during training. JointBal assigns weights proportional to the inverse of the observed joint probabilities of (task score, demographic group), encouraging all (task score, demographic group) combinations to contribute equally for training. Task-JointBal instead uses $\frac{p(y)}{p(y,g)}$, which is equivalent to $\frac{1}{p(g|y)}$, balancing demographic group representation within each score. Conversely, Demo-JointBal uses $\frac{p(g)}{p(y,g)}$, equivalent to $\frac{1}{p(y|g)}$, thereby balancing score distribution within each demographic group. CondDemoBal further refines the idea behind Task-JointBal by weighting samples as $\frac{p(y)}{p(g|y)}$, which ensures that demographic subgroups are equally represented within the same score while still respecting the empirical distribution of scores. DTIndBal assumes the independence between task score and demographic tritute, thus weighting each sample as the ratio of expected probability to the observed probability, encouraging the data to approximate an independence structure.

6. Study setup

6.1. Comparison metrics and debiasing methods

To demonstrate the advantages of the proposed metrics for measuring bias in continuous predictions, this study also includes several state-of-the-art fairness metrics that have been widely used in education, including R^2 -OSA, R^2 -OSD, and R^2 -CSD (Loukina et al., 2019; Yang et al., 2024). Detailed implementation of these metrics and proposed metrics is provided in Appendix C. Additionally, *Mean Difference* is also calculated to compare group differences over the mean of examined outcome values. Furthermore, to evaluate the effectiveness of the proposed

data balancing methods, this study includes two state-of-the-art data debiasing methods for comparisons as introduced below.

- **Demographic Balance Reweighting (DemoRW)** is a simple reweighting strategy (Mohamed & Schuller, 2022) that reweights a sample based on the fraction of its demographic label in the whole dataset, i.e., $w_i = \frac{n}{|G|n_{g_i}}$, where n is the total number of instances n_{g_i} is the number of samples in the group of instance i and $|G|$ is the number of possible values of the demographic label.
- **Relabeling** (Mohamed & Schuller, 2022) transforms the original task scores through a normalization procedure, under the assumption that samples from different demographic groups should share a common mean and standard deviation for task scores, while each sample retains its group-specific z-score. Mathematically, for a sample i , the relabeled task score $\tilde{y}_i$ is computed as $\tilde{y}_i = \frac{y_i - \mu_{g_i}}{\sigma_{g_i}} \cdot \sigma + \mu$, where y_i is the original task score, μ_{g_i} and σ_{g_i} denote the mean and standard deviation of task scores within sample i 's demographic group g_i , respectively, and μ and σ denote the overall mean and standard deviation across all samples.
- **Fair Regularization (FaiReg)** (Bakharia, 2016) is one of the most representative strategies on debias model training by enforcing fairness requirements to a model in the form of regularizers, which penalize a linear regression model for treating similarly scored instances from different demographic groups differently.
- **Reduction** (Agarwal et al., 2019) is one of the most widely cited constraint-based debiasing approaches, which formulates fair regression as a constrained optimization problem by enforcing bounds on group-wise losses, ensuring that the training loss for each demographic subgroup remains below a predefined threshold.

6.2. Implementation of proposed metrics and debiasing methods

AreaCDF assesses distributional differences via the cumulative distribution function (CDF), while *AreaPDF* does so via the probability density function (PDF). We provide robust, ready-to-use implementations of both metrics by adapting well-validated R packages. For *AreaCDF*, we followed a similar approach used in `stats::ecdf()` from the `ggplot2` package (R Core Team & others, 2021; Wickham, 2011), which computes the empirical CDF by sorting the unique observations, accumulating their frequencies, and normalizing by the sample size to obtain the cumulative probability. We then apply `stats::approxfun()` to obtain a continuous, linearly interpolated approximation of the (stepwise) empirical CDF for integration and evaluation. For *AreaPDF*, we estimate densities using kernel density estimation (e.g., `stats::density()` with the default kernel function (Gaussian) and bandwidth selectors (Silverman's rule). For conditional metrics, we excluded bins where any group has fewer than two samples from the conditional metric calculation. This choice prevents high-variance estimates from disproportionately influencing the aggregate metric.

We implemented *Reduction* using the *Fairlearn* (Bird et al., 2020)² library, and *FaiReg* based on the publicly available code from GitHub.³ Note that these methods are limited to linear regression models, as they rely on modifying the model's objective function or imposing constraints during optimization. Such formulations are inherently tied to specific model structures and cannot be easily extended to other model classes. For debiasing methods proposed in this study, we deliberately avoid smoothing and computed the required probabilities empirically from the data, e.g., the marginal probability of observing a specific score y is the proportion of the value occurring in the whole data, i.e., $p(y) = \frac{n_y}{n}$ where n_y is the count of observations equal to y and n is the total sample size. This choice yields weights that are fully transparent and directly reflect the observed data, while also avoiding the need to introduce additional smoothing hyper-parameters. Although our primary analyses rely

on these count-based weights, exploring alternative variants (e.g., KDE-based estimates or binning-based smoothing) represents a promising direction for future work.

6.3. Experimental setup

Through experiments, we aimed to validate the reliability and advantages of our proposed metrics and proposed methods. Towards this goal, beyond reporting point estimates of bias measured by the proposed metrics, we also (1) quantify uncertainty by reporting 95% confidence intervals. Specifically, we adopted the Bias-Corrected and Accelerated (BC_a) bootstrap confidence interval given its ability to adjust both bias and skewness in the bootstrap distribution Efron (1987). This feature is particularly beneficial in our context, where certain demographic groups may contain limited instances, leading to asymmetric or biased resampling distributions. (2) Furthermore, we provide additional supporting evidence for the significance of the observed improvements in predictive fairness. Following the same resampling procedure, we compute the percentage improvement in fairness metrics after applying the proposed debiasing methods and report the corresponding 95% BC_a confidence intervals for these percentage changes. (3) We additionally examine the sensitivity and robustness of our proposed metrics to variations in key parameters, including grouping choices in conditional bias measurement (i.e., *AreaPDF-CSA* and *AreaPDF-CSD*) and bandwidth selection in kernel density estimation used in *AreaPDF*. For grouping choices in *AreaPDF-CSA* and *AreaPDF-CSD*, we investigate the effect of the number of groups by varying the number of bins used to construct conditional distributions (5, 10, 15, and 20), and compare results across different grouping methods, namely *Equal-Width* and *Equal-Frequency*. Specifically, *Equal-Width* partitions students into groups with equal score intervals, whereas *Equal-Frequency* partitions students into groups of equal size. Regarding bandwidth selection, we adopt *Silverman's rule of thumb*, which determines the bandwidth by balancing smoothness to avoid under- or over-fitting based on sample size and deviations. To assess the impact of bandwidth choice on bias measurement, we further evaluate a range of adjustment factors (0.5, 0.75, 1, 1.25, 1.5, and 2), which scale the selected bandwidth by different multiplicative factors. Specifically, we (1) compare bias changes across the two grouping methods, (2) examine how changes in granularity affect metric behavior for students at the extremes of performance, (3) analyze the overall variation in metric values across models as bin granularity increases, and (4) compare how bias changes for both aggregated-level and tail groups under different bandwidth adjustments.

While fairness is the main focus of this study, it is equally important to examine the potential trade-off between accuracy and fairness, as suggested in prior studies (Agarwal et al., 2019; Yang et al., 2024). For predictive accuracy, this study uses Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), where MAE calculates the mean over the absolute values of all prediction errors, and MAPE is calculated as the mean of the absolute percentage errors. The original dataset is split into training and testing sets in an 80:20 ratio, with 80% of the data randomly sampled for training. For all approaches, optimal hyper-parameters are selected through 5-fold cross-validation to ensure the best model performance. The code is publicly available in the GitHub repository⁴(will be available upon acceptance).

7. Results

7.1. Bias evaluation for standardized test score prediction

Table 3 presents bias with respect to gender and disability status across four dimensions (OSA, OSD, CSA, CSD) for numeracy and reading, evaluated on LR, MLP, and XGBoost models using both the prior and proposed metrics. Based on these results, several interesting observations can be made as detailed below.

² https://fairlearn.org/v0.13/api_reference/index.html.

³ <https://github.com/ManishChandra12/Fair-regression>.

⁴ <https://github.com/lilylikeart/Fairness-Continuous-Outcomes>

Table 3

Bias measured with metrics from prior studies and the current study across four dimensions of predicted outcomes (overall score accuracy, overall score difference, conditional score accuracy, and conditional score difference) from three prediction models (LR, MLP, XGBOOST) and actual scores, for the four task-attribute scenarios: (numeracy, reading) $\times$ (gender, disability). The best fairness performance measured by the proposed metrics across the three models is highlighted in bold.

Score outcomes	Source	Metrics	RowID	Prediction model	Task-attribute			
					Numeracy gender	Reading gender	Numeracy disability	Reading disability
Actual Scores	Current Study	AreaCDF	1	–	0.022	0.011	0.041	0.040
		AreaPDF	2	–	0.280	0.137	0.562	0.444
Overall Score Accuracy	Prior Studies	Mean Difference	3	LR	0.004	0.002	0.002	–0.005
			4	MLP	0.004	0.002	0.001	–0.005
			5	XGBOOST	0.001	0.002	0.003	–0.006
		R^2 -OSA	6	LR	ns	ns	ns	ns
			7	MLP	ns	ns	ns	ns
			8	XGBOOST	ns	ns	ns	ns
	Current Study	AreaCDF-OSA	9	LR	0.006 (0.003–0.009)	0.003 (0.002–0.004)	0.003 (0.002–0.003)	0.006 (0.003–0.010)
			10	MLP	0.006 (0.003–0.009)	0.003 (0.002–0.005)	0.002 (0.002–0.002)	0.006 (0.003–0.010)
			11	XGBOOST	0.003 (0.002–0.004)	0.003 (0.002–0.004)	0.004 (0.002–0.005)	0.006 (0.002–0.011)
		AreaPDF-OSA	12	LR	0.275 (0.145–0.385)	0.153 (0.077–0.207)	0.138 (0.076–0.188)	0.274 (0.140–0.412)
			13	MLP	0.279 (0.144–0.400)	0.183 (0.079–0.248)	0.118 (0.073–0.152)	0.307 (0.164–0.444)
			14	XGBOOST	0.139 (0.083–0.181)	0.140 (0.081–0.164)	0.253 (0.108–0.390)	0.230 (0.103–0.364)
Overall Score Difference	Prior Studies	Mean Difference	15	LR	–0.014	0.004	0.000	–0.001
			16	MLP	–0.013	0.005	0.000	–0.001
			17	XGBOOST	–0.016	0.006	0.001	0.000
		R^2 -OSD	18	LR	0.021	ns	ns	ns
			19	MLP	0.019	ns	ns	ns
			20	XGBOOST	0.027	ns	ns	ns
	Current Study	AreaCDF-OSD	21	LR	0.014 (0.008–0.023)	0.007 (0.004–0.011)	0.003 (0.003–0.003)	0.006 (0.004–0.007)
			22	MLP	0.013 (0.008–0.021)	0.007 (0.004–0.011)	0.003 (0.004–0.004)	0.006 (0.004–0.009)
			23	XGBOOST	0.016 (0.008–0.025)	0.008 (0.004–0.014)	0.004 (0.003–0.004)	0.007 (0.003–0.010)
		AreaPDF-OSD	24	LR	0.299 (0.172–0.391)	0.156 (0.099–0.190)	0.111 (0.085–0.113)	0.206 (0.093–0.292)
			25	MLP	0.294 (0.171–0.382)	0.158 (0.097–0.187)	0.103 (0.069–0.097)	0.230 (0.099–0.331)
			26	XGBOOST	0.232 (0.117–0.333)	0.176 (0.122–0.219)	0.181 (0.089–0.246)	0.249 (0.109–0.345)
Conditional Score Accuracy	Current Study	AreaPDF-CSA	27	LR	0.514 (0.356–0.586)	0.512 (0.388–0.558)	0.646 (0.432–0.725)	0.498 (0.444–0.477)
			28	MLP	0.501 (0.376–0.537)	0.544 (0.421–0.583)	0.606 (0.426–0.667)	0.560 (0.431–0.600)
			29	XGBOOST	0.427 (0.345–0.440)	0.479 (0.345–0.542)	0.750 (0.517–0.890)	0.489 (0.458–0.463)
Conditional Score Difference	Prior Studies	R^2 -CSD	30	LR	ns	ns	0.035	0.014
			31	MLP	ns	ns	0.034	0.013
			32	XGBOOST	ns	ns	0.037	0.020
	Current Study	AreaPDF-CSD	33	LR	0.514 (0.369–0.590)	0.492 (0.399–0.515)	0.714 (0.513–0.809)	0.537 (0.490–0.494)
			34	MLP	0.505 (0.368–0.568)	0.456 (0.373–0.462)	0.670 (0.464–0.746)	0.549 (0.501–0.541)
			35	XGBOOST	0.428 (0.319–0.478)	0.519 (0.391–0.586)	0.772 (0.572–0.859)	0.535 (0.473–0.489)

Regarding **bias detected through prior metrics vs proposed metrics**, it can be observed that, within each of the three dimensions of predicted outcomes investigated in prior studies (i.e., OSA (Rows 3–14), OSD (Rows 15–26), and CSD (Rows 30–35)), all metrics — whether from prior studies or the current study — consistently identify the same scenario as more likely exhibiting the highest bias among the four Task-Attribute scenarios. This consistency highlights the effectiveness of these metrics in identifying the highest bias scenarios and their alignment in relative comparisons across different cases. For example, Numeracy-Gender was consistently identified as the scenario with the largest bias in OSD by all metrics (Rows 15–26), indicating that predicted numeracy test scores differ most significantly between male and female students when compared to the actual scores. However, using R^2 -based metrics, only 9 cases out of the 36 were found to exhibit bias: R^2 -OSD for Numeracy-Gender and R^2 -CSD for Numeracy-Disability and for Reading-Disability across the three prediction models. All other cases showed non-significant results, with none exhibiting significant effects of demographic groups using R^2 -OSA (Rows 6–8). Based on these results, a conclusion could be drawn that the predictions were not biased for those “ns” cases. However, the proposed metrics in this

study, i.e., *AreaCDF-OSA*, *AreaPDF-OSA*, *AreaCDF-OSD*, and *AreaPDF-OSD*, reveal that disparities still exist between groups in these cases to varying degrees. In particular, it's obvious that for OSA different degrees of bias exist for different scenarios according to *AreaCDF-OSA* and *AreaPDF-OSA*, but R^2 -OSA fails to identify any demographic effects.

From Table 3 it can also be observed **how bias manifests differently across different dimensions**. Comparing *AreaPDF-CSA* (Rows 27–29) with *AreaPDF-OSA* (Rows 12–14) and *AreaPDF-CSD* (Rows 33–35) with *AreaPDF-OSD* (Rows 24–26) within the same Task-Attribute scenario, it can be seen that disparities in score accuracy and score differences between groups become significantly more pronounced when further considering students with similar actual competence. Particularly, for numeracy predictions between students with disabilities and those without, the predictive bias of LR observed in OSA (*AreaPDF-OSA*: 0.138 in Row 12) and OSD (*AreaPDF-OSD*: 0.111 in Row 24) is relatively small compared to other cases. However, when accounting for students' actual numeracy competency, significantly higher biases emerge, as indicated by the much larger values of *AreaPDF-CSA* (0.646 in Row 27) and *AreaPDF-CSD* (0.714 in Row 33), which are the highest among the four

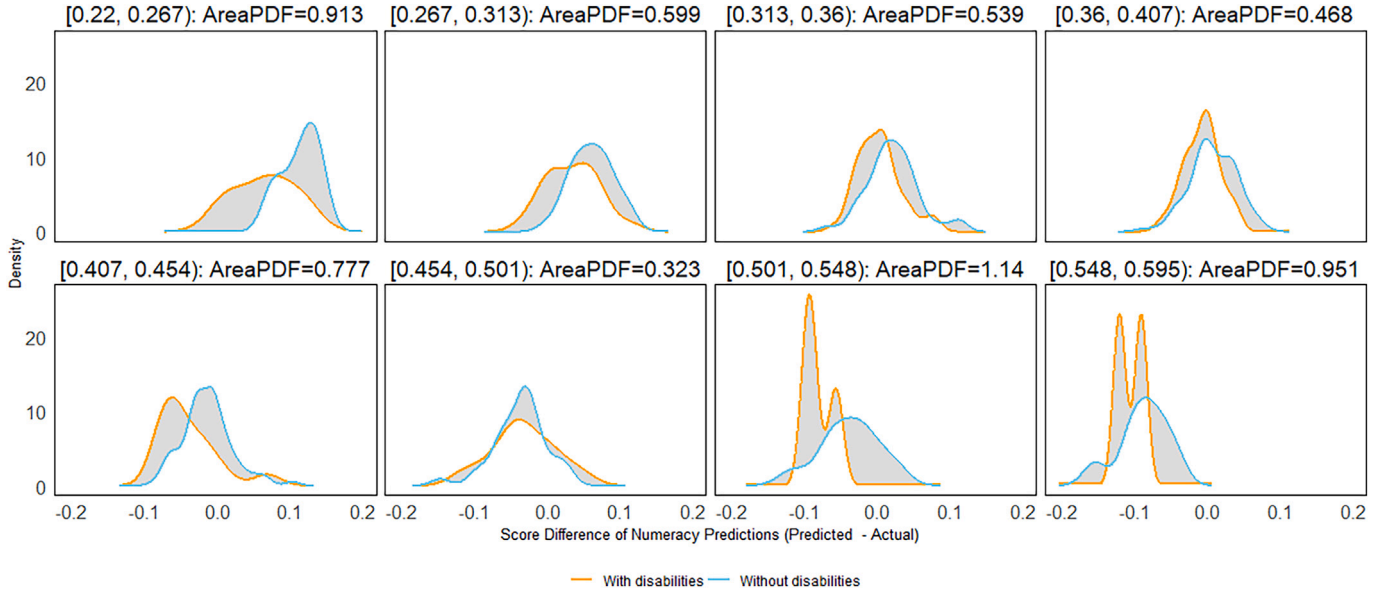


Fig. 2. Illustration of bias in the conditional score difference of numeracy predictions between students with disabilities and those without. Above each subplot the actual score range and corresponding *AreaPDF-OSD* are provided. *AreaPDF-CSD* is calculated as the average over all these values. Values over the x-axis are prediction errors, i.e., predicted scores - actual scores.

cases. This highlights a more concerning issue: a considerable number of students with similar actual competence but different disability statuses receive vastly different predicted scores, suggesting that biases remain hidden when actual competence is not explicitly considered. Further breaking down *AreaPDF-CSD* calculated on LR predictions for numeracy-disability as shown in Fig. 2, higher biases were observed among students with exceptional (the last two sub-figures in the second row) or very low actual test performance (the first sub-figure). In these cases, students without disabilities are more likely to be overestimated, receiving higher predicted scores than students with disabilities, despite both groups having lower actual competency. This reveals that students at both ends of the performance spectrum, i.e., those who excel and those who struggle, are more susceptible to biases related to their demographic background.

In terms of **bias exhibited by different predictive models**, we observed that model behavior varies substantially across scenarios. For instance, in the case of numeracy predictions stratified by disability status, the MLP model demonstrates the lowest level of bias across all score outcomes (see Rows 10, 13, 22, 25, 28 & 34), whereas for numeracy predictions by gender, XGBoost consistently produces the smallest bias across most score outcomes, such as OSA (Rows 11 & 14), OSD (Row 26), CSA (Row 29) and CSD (Row 35). These results suggest that different models respond differently to subgroup distributions, which may be partially attributed to their model design, particularly how each model captures and weights feature interactions.

Together with Table 1, Table 3 shows that although male and female students are nearly equally represented in the training data (ratio of 1.03 : 1), this demographic balance does not necessarily translate to greater fairness in OSA (Rows 3–14) and OSD (Rows 15–26). For example, bias in LR predictions observed for the Numeracy-Gender scenario is largest over OSA and OSD among all the four task-attribute scenarios (as visualized in Fig. D.2 in Appendix D). In contrast, for the same set of numeracy predictions, bias related to disability status is the lowest regarding OSA and OSD despite the disability attribute being more imbalanced. This observation underscores that demographic balance alone does not guarantee fairness in continuous predictive outcomes, which has been similarly identified for binary prediction tasks (Li et al., 2025). Furthermore, Rows 1 and 2 of Table 3, along with Fig. D.1

in Appendix D, show that students with disabilities scored substantially lower than their peers in both numeracy and reading. Interestingly, despite the large differences in actual score distributions between students with disabilities and those without, the corresponding biases of the three predictive models in OSA and OSD are not necessarily the highest among the task-attribute scenarios, whereas CSA and CSD exhibit the greatest bias. This suggests that disparities in score distributions between groups might not necessarily result in higher bias in OSA and OSD, they may contribute to greater bias when comparing students with similar actual competence.

7.2. Sensitivity of *AreaPDF-CSA* and *AreaPDF-CSD* under different grouping choices

In this section, we present results for *AreaPDF-CSA* and *AreaPDF-CSD* under different grouping choices, focusing on two factors: (i) the grouping method and (ii) the number of groups. For grouping methods, we compare *Equal-Width* and *Equal-Frequency binning*, while for the number of groups we consider the set {5, 10, 15, 20}.

7.2.1. Effects of grouping methods: equal-width vs. equal-frequency

From Fig. 3, we observe that in most cases the two grouping methods, i.e., *Equal-Width* and *Equal-Frequency binning*, produce very similar values when measuring bias with *AreaPDF-CSA* and *AreaPDF-CSD*, particularly in the Numeracy-Gender and Reading-Disability scenarios. In the Reading-Gender and Numeracy-Disability scenarios, the values obtained using equal-frequency binning are smaller than those obtained using equal-width binning. This suggests that although grouping choice can influence the magnitude of the bias detected, the bias is largely driven by broad distributional differences rather than the specific partitioning of the score axis. However, noticeable differences may emerge when the two grouping methods produce markedly different score ranges in the tail regions. In such cases, under *Equal-Frequency binning*, the highest bin may still contain substantial group differences, but these differences are compressed into a single bin. Consequently, their contribution to conditional bias is limited to that single interval. In contrast, *Equal-Width binning* often produces several bins in the upper tail. When group distributions diverge in this region, those differences are spread

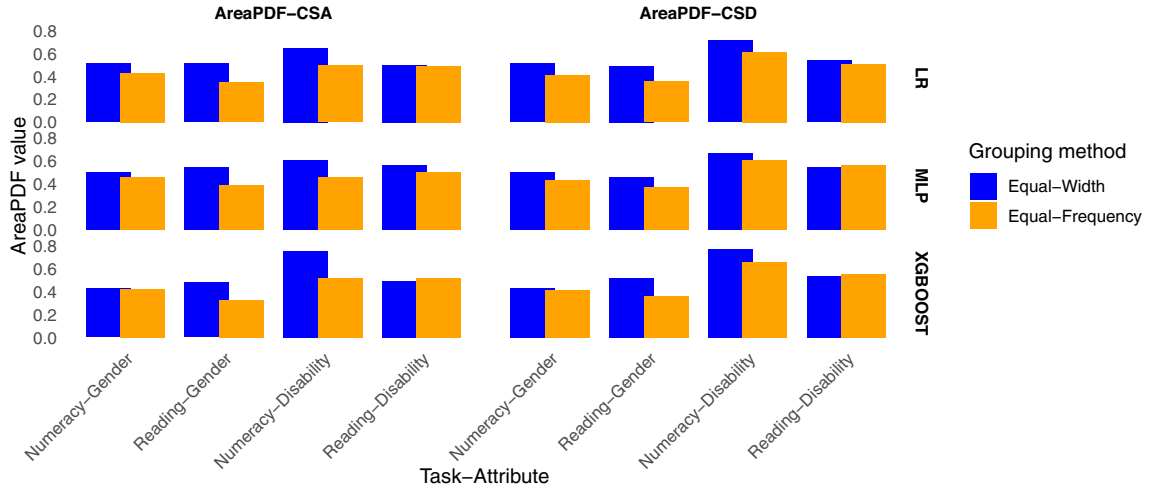


Fig. 3. Illustration of **conditional bias in score accuracy and difference** measured by *AreaPDF-CSA* and *AreaPDF-CSD* across four task-attribute scenarios (numeracy-gender, reading-gender, numeracy-disability, and reading-disability), and three prediction models (LR, MLP and XGBOOST). Results are shown for two grouping methods: *Equal-Width* and *Equal-Frequency* binning.

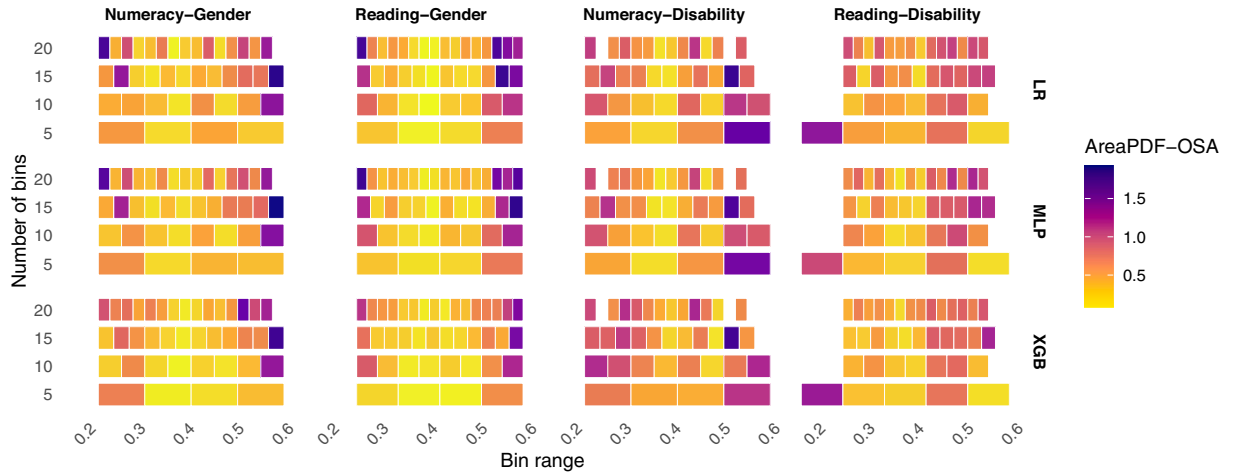


Fig. 4. Illustration of **conditional bias in score accuracy** measured by *AreaPDF-CSA*, stratified by conditional groupings over ground-truth scores across four task-attribute scenarios (numeracy-gender, reading-gender, numeracy-disability, and reading-disability), and three prediction models (LR, MLP and XGBOOST). The x-axis represents the score range per bin, while the y-axis denotes different bin-size configurations (5, 10, 15, and 20). Darker colors indicate higher levels of bias. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

across multiple bins, each contributing to the bias measure, which can make the overall conditional bias appear larger.

7.2.2. Effect of the number of groups

For conditional metrics i.e., *AreaPDF-CSA* and *AreaPDF-CSD*, (1) to illustrate how changes in granularity influence metric behavior for students at the extremes of performance, we computed the AreaPDF for each bin under different bin settings (5, 10, 15, and 20) and visualized the results in Figs. 4 and 5. (2) Additionally, to examine how the overall values of these two metrics change across different levels of bin granularity, we visualized the aggregated results in Fig. 6, which presents AreaPDF-CSA and AreaPDF-CSD values computed by averaging the AreaPDF scores across all bins.

As shown in Figs. 4 and 5, across different numbers of bins and prediction models, the highest levels of bias in score accuracy and score difference were consistently observed at the two extremes of performance. For example, in the numeracy-disability scenario, the right end of the distribution, corresponding to high-performing students, tends to exhibit the greatest bias across all bin configurations. This

pattern indicates that fairness disparities are most pronounced among students at both the lower and higher ends of the performance spectrum. For both metrics, we also observed that in the reading-disability scenario, only the 5-bin configuration shows a noticeable left-end bias, whereas the remaining configurations appear empty. This likely occurs because, as the number of bins increases, certain bins may contain data points from only one subgroup or only one point included, making the calculation of the conditional metric infeasible or unstable for those bins.

From Fig. 6, we can observe that for the same prediction model under a given task-attribute scenario, both metrics exhibit similar patterns of value change across different bin configurations. For example, across all four task-attribute scenarios, the lines with the same color but different shapes (e.g., LR shown in red, where the solid line represents AreaPDF-CSA and the dashed line corresponds to AreaPDF-CSD) vary in a similar manner, indicating consistent metric behavior across different levels of granularity. Moreover, the overall trends of metric variation across models remain largely consistent within the same demographic attribute. For instance, under the disability attribute, both numeracy and

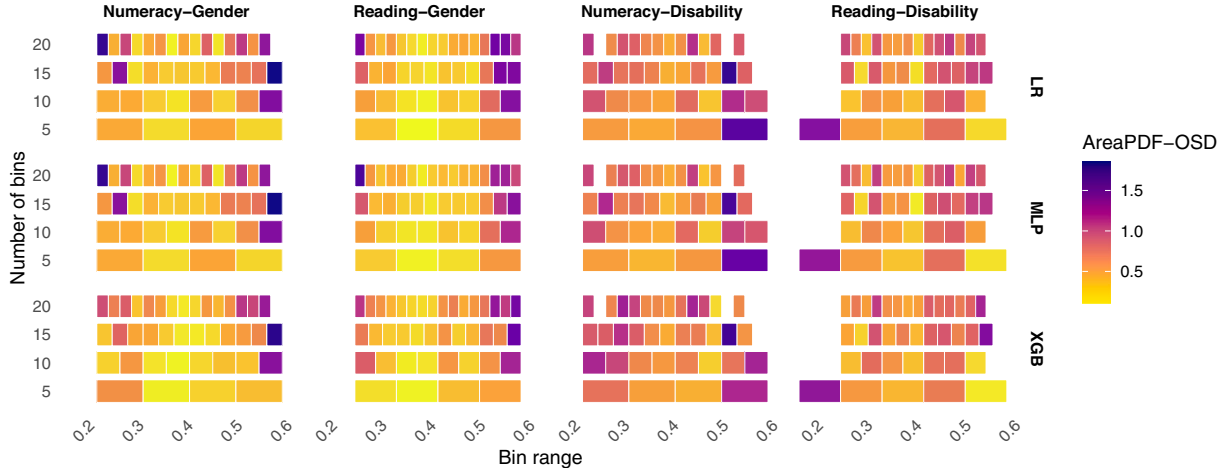


Fig. 5. Illustration of **conditional bias in score difference** measured by *AreaPDF-CSD*, stratified by conditional groupings over ground-truth scores across four task-attribute scenarios (numeracy-gender, reading-gender, numeracy-disability, and reading-disability) and three prediction models (LR, MLP and XGBOOST). The x-axis represents the score range per bin, while the y-axis denotes different bin-size configurations (5, 10, 15, and 20). Darker colors indicate higher levels of bias. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

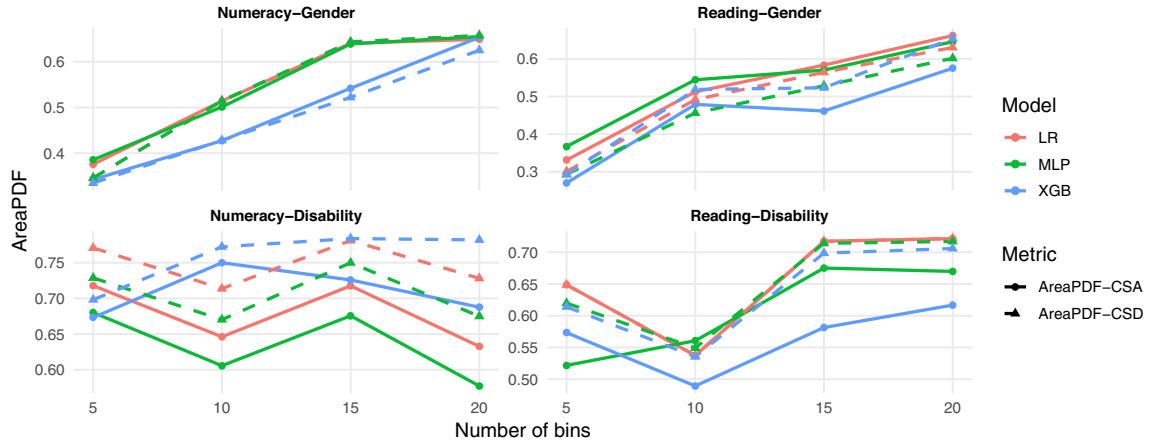


Fig. 6. Illustration of **conditional bias in score accuracy and score difference** measured by *AreaPDF-CSA* and *AreaPDF-CSD* under different bin-size configurations (5, 10, 15, and 20) across four task-attribute scenarios (numeracy-gender, reading-gender, numeracy-disability, and reading-disability) and three prediction models (LR, MLP, and XGBoost). The x-axis represents the bin size, while the y-axis indicates the corresponding bias value. Different colors denote prediction models, and line styles distinguish the metrics, solid line for *AreaPDF-CSA* and dashed line for *AreaPDF-CSD*. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

reading predictions display similar patterns, where a bin size of 10 tends to yield smaller bias compared to other bin configurations. It can also be seen that, in the reading-disability scenario, the 5-bin configuration tends to produce higher bias than the 10-bin case, which might be due to the finer granularity causing some bins to be excluded from the metric calculation. From another perspective, this bin-size analysis can also serve as a diagnostic tool for detecting the presence of extreme bins or sparsely populated regions in the data.

7.2.3. Sensitivity of bandwidth selection in density estimation

Figs. 7 and 8 present the sensitivity of *AreaPDF* computed with different bandwidth selections in kernel density estimation, for both aggregated conditional bias (*AreaPDF-CSA* and *AreaPDF-CSD*) and tail-specific bias (*AreaPDF-OSA* and *AreaPDF-OSD*), respectively. Using the unadjusted bandwidth (i.e., adjustment = 1) as the baseline, we observe that the relative ranking of biases across task-attribute scenarios remains consistent, with most variations within 10% of the baseline. Notably, when focusing on tail score groups (i.e., students in the lowest and highest score bins), the changes in bias measured by *AreaPDF*

across different bandwidth selections are even smaller across all scenarios, further indicating the stability of the proposed metrics with respect to bandwidth choice.

7.3. Effectiveness of data balancing in improving predictive fairness

Figs. 9–11 present the predictive fairness improvements for numeracy and reading outcomes obtained using LR, MLP, and XGBoost, respectively, after applying a range of data debiasing methods, including both existing approaches and our proposed methods, and Table 4 provides the predictive accuracy measured by MAE and MAPE after applying these methods. Detailed values for debiasing performance can also be found in Tables D.1–D.3 in Appendix D.

7.3.1. Predictive fairness

From Figs. 9–11 and results in Tables D.1–D.3, several important observations can be obtained.

Overall Effectiveness of Proposed Debiasing Methods. With the six fairness metrics (i.e., *AreaCDF-OSA*, *AreaPDF-OSA*, *AreaCDF-OSD*, *AreaPDF-OSD*, *AreaPDF-CSA*, and *AreaPDF-CSD*) applied to each of the

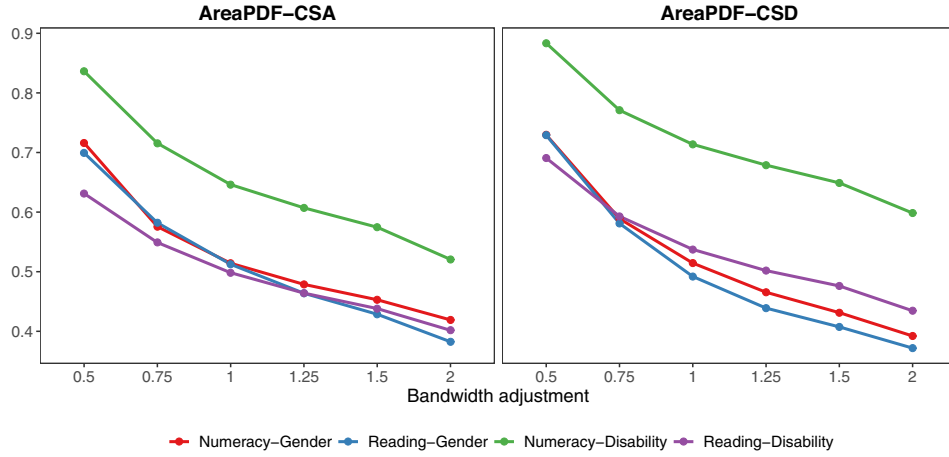


Fig. 7. Illustration of **conditional bias in score accuracy and score difference** for LR predictions measured by *AreaPDF-CSA* and *AreaPDF-CSD*. Results are shown across varying KDE bandwidth adjustments under four task-attribute scenarios: numeracy-gender, reading-gender, numeracy-disability, and reading-disability. The x-axis represents the bandwidth adjustment, and the y-axis denotes the corresponding bias value.

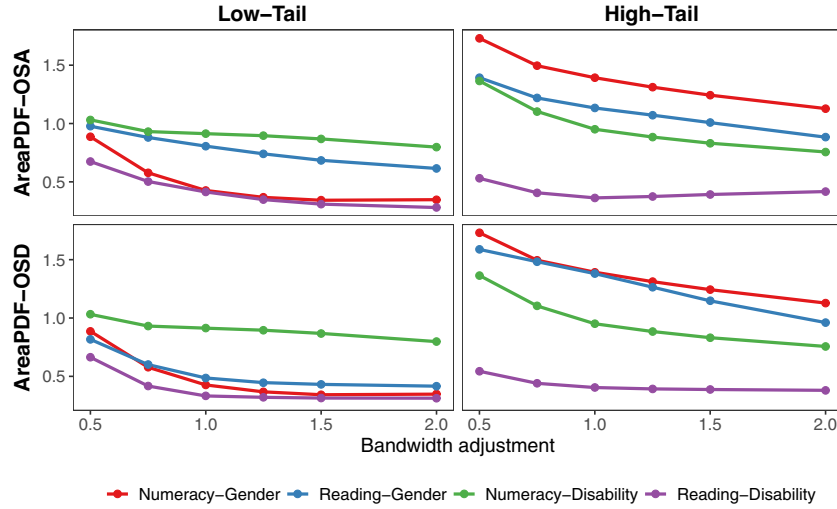


Fig. 8. Illustration of **bias in score accuracy and score difference** for LR predictions at the two distributional tails (lowest and highest score bins), measured using *AreaPDF-OSA* and *AreaPDF-OSD*. Results are shown across varying KDE bandwidth adjustments under four task-attribute scenarios: numeracy-gender, reading-gender, numeracy-disability, and reading-disability. The x-axis represents the bandwidth adjustment, and the y-axis denotes the corresponding bias value.

four task-attribute scenarios (i.e., Numeracy-Gender, Reading-Gender, Numeracy-Disability, and Reading-Disability), each method produced 24 outcomes quantifying the extent to which the predictions of a given model exhibit bias. From the three bar charts in Figs. 9–11, it can be seen that, the proposed methods achieved fairness improvements on average in 12 out of 24 cases across the three prediction models, whereas the baseline methods, i.e., DemoRW and Relabel, showed improvements in only 6 and 9 cases out of 24 on average, respectively. In particular, both in-processing methods, FairReg and Reduction, improved fairness in only 7 out of 24 cases and achieved the best fairness performance in just 1 out of 24 cases, showing limited effectiveness in this setting. Among all debiasing methods, CondDemoBal consistently outperformed others, ranking among the top two best-performing approaches across all models and achieving the greatest improvements in at least 8 out of 24 cases for each predictive model. Following that, Demo-JointBal achieved the best fairness improvements in 7, 2, and 3 cases out of 24 for LR, MLP, and XGBoost, respectively. These results indicate that the five proposed reweighting methods were generally more effective in enhancing predictive fairness across diverse metrics, models, and task-attribute scenarios compared to the existing baselines.

Diverse Patterns of Effectiveness of Proposed Debiasing Methods across Different Predictive Models. Beyond the overall fairness improvements achieved by the proposed reweighting methods, the effectiveness of individual debiasing strategies varies notably across models and prediction scenarios. For instance, while DemoRW exhibits very limited effectiveness when applied to LR predictions, it yields considerably better fairness outcomes for MLP and XGBoost predictions, with a larger number of cases showing significant bias reductions.

- For LR predictions, all proposed reweighting-based methods substantially improved fairness for predictions stratified by the gender attribute, compared with DemoRW and Relabeling. As shown in Fig. 9, most cells corresponding to Task-JointBal, CondDemo-JointBal, and DTIndBal in the Numeracy-Gender and Reading-Gender settings are shaded green, indicating reduced bias. Consistent with this pattern, Table D.1 (Rows 7–12) shows that the magnitude of reduction ranges from 3.7% to 42.9%, with nearly all cases exhibiting confidence intervals that exclude zero, suggesting statistically meaningful fairness improvements.

Table 4

Predictive accuracy measured by MAE and MAPE, before and after applying existing data balancing methods and the proposed methods across the four task–attribute scenarios. Values in parentheses indicate the percentage change in accuracy relative to the unbiased setting, with negative values representing reduced accuracy. Reductions greater than 5% are highlighted in bold.

Metric	Task-attribute	RowID	Prediction model	Unbiased	Previous methods		Proposed methods				
					DemoRW	Relabeling	JointBal	Task-JointBal	Demo-JointBal	CondDemoBal	DTIndBal
MAE	Numeracy-Gender	1	LR	0.035	0.035 (0.0%)	0.036 (−2.9%)	0.036 (−2.9%)	0.036 (−2.9%)	0.036 (−2.9%)	0.037 (−5.7%)	0.036 (−2.9%)
		2	MLP	0.035	0.037 (−5.7%)	0.036 (−2.9%)	0.036 (−2.9%)	0.036 (−2.9%)	0.036 (−2.9%)	0.055 (−57.1%)	0.036 (−2.9%)
		3	XGBOOST	0.036	0.036 (0.0%)	0.037 (−2.8%)	0.039 (−8.3%)	0.038 (−5.6%)	0.039 (−8.3%)	0.055 (−52.8%)	0.036 (0.0%)
	Reading-Gender	4	LR	0.035	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.034 (2.9%)	0.033 (5.7%)
		5	MLP	0.035	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)	0.033 (5.7%)
		6	XGBOOST	0.036	0.031 (13.9%)	0.032 (11.1%)	0.034 (5.6%)	0.032 (11.1%)	0.033 (8.3%)	0.061 (−69.4%)	0.031 (13.9%)
	Numeracy-Disability	7	LR	0.036	0.035 (2.8%)	0.036 (0.0%)	0.036 (0.0%)	0.036 (0.0%)	0.036 (0.0%)	0.037 (−2.8%)	0.036 (0.0%)
		8	MLP	0.036	0.036 (0.0%)	0.036 (0.0%)	0.036 (0.0%)	0.036 (0.0%)	0.036 (0.0%)	0.055 (−52.8%)	0.036 (0.0%)
		9	XGBOOST	0.036	0.037 (−2.8%)	0.038 (−5.6%)	0.039 (−8.3%)	0.037 (−2.8%)	0.041 (−13.9%)	0.055 (−52.8%)	0.038 (−5.6%)
	Reading-Disability	10	LR	0.036	0.033 (8.3%)	0.034 (5.6%)	0.033 (8.3%)	0.033 (8.3%)	0.033 (8.3%)	0.035 (2.8%)	0.033 (8.3%)
		11	MLP	0.037	0.033 (10.8%)	0.034 (8.1%)	0.033 (10.8%)	0.033 (10.8%)	0.033 (10.8%)	0.034 (8.1%)	0.033 (10.8%)
		12	XGBOOST	0.037	0.032 (13.5%)	0.033 (10.8%)	0.033 (10.8%)	0.031 (16.2%)	0.032 (13.5%)	0.061 (−64.9%)	0.031 (16.2%)
MAPE	Numeracy-Gender	13	LR	0.095	0.095 (0.0%)	0.095 (0.0%)	0.095 (0.0%)	0.095 (0.0%)	0.095 (0.0%)	0.099 (−4.2%)	0.095 (0.0%)
		14	MLP	0.095	0.099 (−4.2%)	0.096 (−1.1%)	0.097 (−2.1%)	0.097 (−2.1%)	0.096 (−1.1%)	0.145 (−52.6%)	0.095 (0.0%)
		15	XGBOOST	0.098	0.097 (1.0%)	0.100 (−2.0%)	0.104 (−6.1%)	0.101 (−3.1%)	0.103 (−5.1%)	0.145 (−48.0%)	0.096 (2.0%)
	Reading-Gender	16	LR	0.088	0.088 (0.0%)	0.089 (−1.1%)	0.088 (0.0%)	0.089 (−1.1%)	0.088 (0.0%)	0.092 (−4.5%)	0.089 (−1.1%)
		17	MLP	0.088	0.088 (0.0%)	0.089 (−1.1%)	0.089 (−1.1%)	0.089 (−1.1%)	0.089 (−1.1%)	0.091 (−3.4%)	0.089 (−1.1%)
		18	XGBOOST	0.084	0.085 (−1.2%)	0.086 (−2.4%)	0.092 (−9.5%)	0.087 (−3.6%)	0.089 (−6.0%)	0.158 (−88.1%)	0.085 (−1.2%)
	Numeracy-Disability	19	LR	0.095	0.095 (0.0%)	0.096 (−1.1%)	0.096 (−1.1%)	0.096 (−1.1%)	0.097 (−2.1%)	0.099 (−4.2%)	0.096 (−1.1%)
		20	MLP	0.095	0.096 (−1.1%)	0.098 (−3.2%)	0.096 (−1.1%)	0.097 (−2.1%)	0.098 (−3.2%)	0.144 (−51.6%)	0.096 (−1.1%)
		21	XGBOOST	0.098	0.099 (−1.0%)	0.101 (−3.1%)	0.104 (−6.1%)	0.100 (−2.0%)	0.109 (−11.2%)	0.144 (−46.9%)	0.101 (−3.1%)
	Reading-Disability	22	LR	0.088	0.088 (0.0%)	0.092 (−4.5%)	0.088 (0.0%)	0.088 (0.0%)	0.088 (0.0%)	0.093 (−5.7%)	0.088 (0.0%)
		23	MLP	0.088	0.088 (0.0%)	0.091 (−3.4%)	0.089 (−1.1%)	0.088 (0.0%)	0.088 (0.0%)	0.093 (−5.7%)	0.088 (0.0%)
		24	XGBOOST	0.084	0.086 (−2.4%)	0.090 (−7.1%)	0.087 (−3.6%)	0.084 (0.0%)	0.088 (−4.8%)	0.159 (−89.3%)	0.086 (−2.4%)

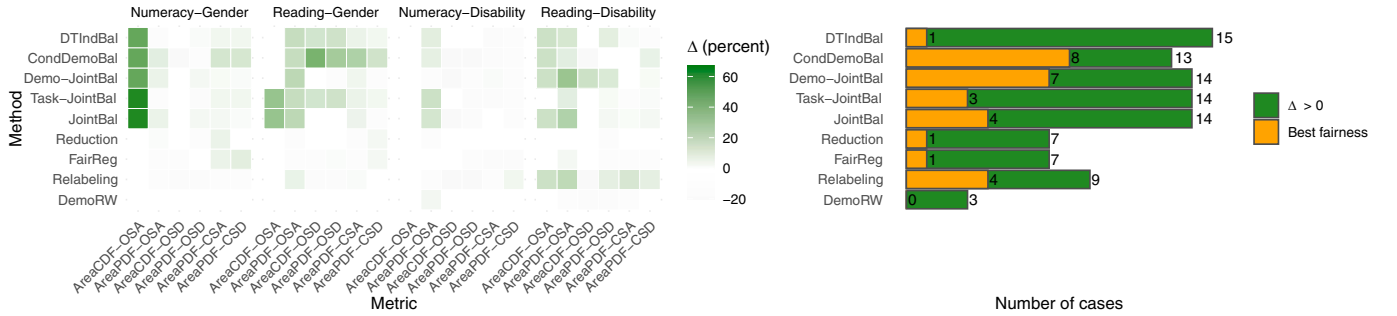


Fig. 9. Fairness performance of all debiasing methods when predicting with LR. The left heatmap shows the percentage reduction in bias, relative to the unbiased baseline, i.e., LR, for each debiasing method across all Task-Attribute scenarios and metrics. The right bar plot summarizes, for each method, the Total number of cases in which bias was reduced ($\Delta > 0$) and achieved the best reduction.

Regarding disability-related bias, although fewer cases demonstrated improvements compared with the gender attribute, the proposed reweighting methods still showed a strong capacity for mitigating bias. For example, in the reading–disability scenario, Demo-JointBal reduced bias across five out of six metrics, achieving the largest reduction of 32.5% in AreaPDF-OSA. It is worth noting that although some confidence intervals included zero, their lower bounds were small (albeit negative), suggesting that the proposed reweighting approaches consistently shifted the estimates toward fairness improvement even when the statistical evidence was marginal. In rare cases where bias slightly increased, the magnitude of change was minimal and remained within an acceptable tolerance (no more than 6.5%).

- For MLP predictions, the proposed reweighting methods exhibited markedly different levels of effectiveness across task–attribute scenarios. For instance, in the Numeracy–Gender scenario, JointBal emerged as the best-performing method, significantly reducing bias across five out of six metrics (corresponding to five green-shared cells in Fig. 10 for Numeracy-Gender), with three of them supported by confidence intervals that provide strong statistical evidence of fairness improvement (see Table D.2). Similarly, in the reading–gender scenario, CondDemoBal demonstrated the best performance. In the numeracy–disability scenario, although all methods were generally less effective, each of the proposed reweighting methods (except DTIndBal) still reduced bias in at least two metrics, whereas neither DemoRW nor Relabeling

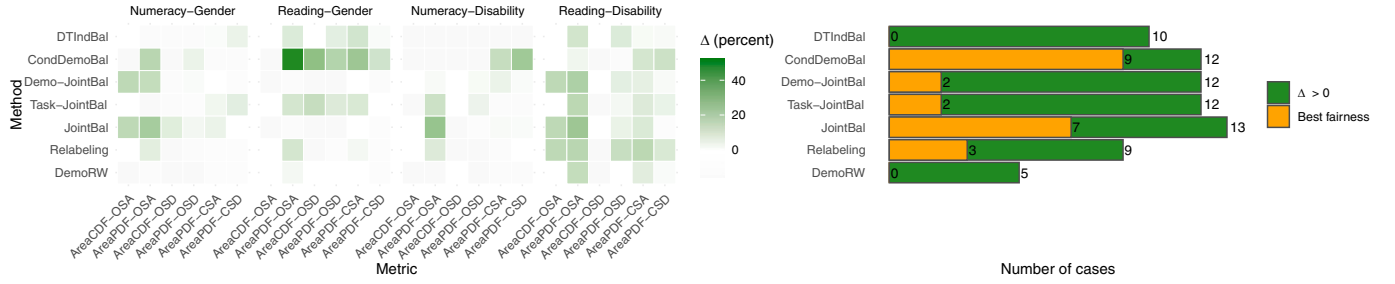


Fig. 10. Fairness performance of all debiasing methods when predicting with MLP. The left heatmap shows the percentage reduction in bias, relative to the unbiased baseline, i.e., MLP, for each debiasing method across all Task-Attribute scenarios and metrics. The right bar plot summarizes, for each method, the Total number of cases in which bias was reduced ($\Delta > 0$).

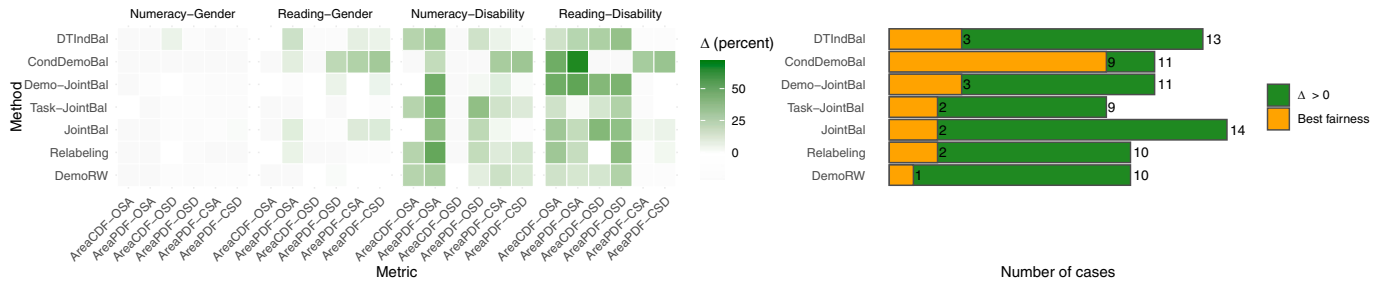


Fig. 11. Fairness performance of all debiasing methods when predicting with XGBoost. The left heatmap shows the percentage reduction in bias, relative to the unbiased baseline, i.e., XGBoost, for each debiasing method across all Task-Attribute scenarios and metrics. The right bar plot summarizes, for each method, the Total number of cases in which bias was reduced ($\Delta > 0$).

achieved any improvement. Finally, in the reading-disability scenario, the proposed reweighting methods consistently yielded the best fairness-improving results, achieving statistical significance for nearly all metrics.

- For XGBoost predictions in Fig. 11 and Table D.3, a clear pattern emerges: all debiasing methods appeared less effective overall in mitigating bias related to gender compared with disability. In the numeracy-gender scenario, only JointBal shows a significant bias reduction by 6.3% (See Numeracy-Gender panel in heatmap of Fig. 11 and Row 6 in Table D.3). The performance on reading predictions was comparatively better, where CondDemoBal achieved the most substantial fairness improvements, reducing bias across three out of six metrics with reductions of at least 22.2%. Another notable observation is that, for disability-related bias, most debiasing methods appeared effective, with statistical significance supported by the corresponding confidence intervals. In particular, nearly all of the best-performing fairness-improving methods with significant evidence fell within our proposed reweighting framework, such as Demo-JointBal and CondDemoBal.

Overall, we noticed that compared to MLP and XGBoost, LR tends to have a higher sensitivity to all proposed reweighting approaches that incorporate finer-grained variations in the sample distribution, than to methods like DemoRW, which operate only at the coarse group-membership level. Moreover, different predictive models appear to be sensitive to the proposed reweighting approaches under different conditions. For example, the overall fairness improvements achieved for LR predictions tend to be more substantial for demographic attributes with relatively balanced representation (e.g., gender) than for those with pronounced imbalance (e.g., disability status). In contrast, for XGBoost predictions, the proposed reweighting methods tend to be more effective in mitigating bias related to the disability attribute. This

variation in model sensitivity may stem from differences in model design, particularly in how features are combined to make predictions. Simpler models such as LR are more directly influenced by distributional shifts introduced through reweighting, whereas more complex models like XGBoost can better adapt to localized adjustments within underrepresented subgroups.

Cross-Model Consistency of Debiasing Methods. Despite the varying patterns observed across models and task-attribute scenarios, some consistent trends can still be identified regarding the relative effectiveness of the proposed methods. Across all three predictive models (LR, MLP, and XGBoost), CondDemoBal demonstrated consistently higher effectiveness in gender-related scenarios, showing improved fairness in 7 out of 12 cases on average and achieving the best fairness improvements in 5 out of 12 cases. In contrast, DemoJointBal tended to perform better in disability-related scenarios, with fairness improvements observed in an average of 8 out of 12 cases and the best fairness outcomes achieved in 3 out of 12 cases. This observation aligns with the design of the two methods: DemoJointBal explicitly accounts for the broader effects of demographic imbalances, whereas CondDemoBal focuses on mitigating localized disparities in task-score distributions within demographic subgroups.

7.3.2. Fairness-accuracy trade-off

From the results in Table 4, it is evident that, in most cases, applying debiasing methods to different predictive models does not necessarily lead to a significant reduction in predictive accuracy. In some instances, accuracy is even further improved following debiasing. For example, in the reading prediction tasks, almost all debiasing methods increased accuracy measured by MAE by approximately 3% to 16%. In contrast, for numeracy predictions, no accuracy gains were observed; however, the accuracy losses in most cases were limited to within 6%. Exceptions did exist, for instance, CondDemoBal, when applied to MLP and XGBoost, resulted in a substantial accuracy

decrease of around 50% in terms of MAE. When connecting this observation with the results in Tables D.2 and D.3, it becomes evident that, for these two predictive models, CondDemoBa1 was also less effective in reducing bias for numeracy predictions compared with reading predictions. This may be attributed to the data composition, where numeracy-oriented features are relatively limited (only four specific to numeracy), while most features are reading-oriented. As a result, reweighting may have disproportionately affected the smaller set of numeracy predictors, amplifying instability and leading to both weaker fairness gains and greater accuracy degradation.

8. Discussion

While it is important to avoid over-generalizing the results of the examined metrics in this specific test score prediction task, the findings highlight that, in general, accounting for the full distribution, using the proposed metrics *AreaCDF* and *AreaPDF*, provides valuable insights into predictive performance disparities that extend beyond simple central tendency comparisons offered by existing metrics like R^2 -OSA. Similar to the spirit of the well-established fairness metric for classification tasks in education, i.e., ABROCA (Gardner et al., 2019), which considers the absolute difference between the ROC curves of two groups at each possible threshold, the proposed metrics *AreaCDF* and *AreaPDF* account for the absolute difference between CDFs or PDFs of two groups at each possible outcome value of predictions. In educational contexts, this characteristic is particularly important, as it helps uncover disparities not only among students with average performance but also among those with exceptionally high or low outcomes, i.e., disparities that are often “cancelled out” or overlooked by metrics relying solely on central tendency. This ensures that fairness evaluations are informed not only by overall trends but also by the experiences of students at both ends of the performance spectrum, i.e., groups that may be especially vulnerable to misprediction and inequitable educational opportunities. For example, as illustrated in Fig. 2, students with very low or very high actual scores exhibit more pronounced disparities in prediction errors across groups, highlighting biases that remain unseen when average performance alone is considered.

Additionally, due to the nature of their statistical distributional distance foundations, both metrics guarantee a robust and versatile evaluation of disparities. For example, unlike R^2 -based metrics that may yield “ns” (non-significant results) and falsely suggest the absence of bias, *AreaCDF* and *AreaPDF* consistently provide a bounded quantitative measure of the distributional differences between groups for the intended outcome. This ensures that disparities are reliably captured, regardless of the specific predictive task or dataset. Furthermore, this property facilitates the comparative analysis of bias across different dimensions of predictive outcomes, particularly in the comparisons of OSA vs. CSA and OSD vs. CSD. As the results show, students with similar competence levels but from different demographic groups may experience significantly different predictions, even when the overall accuracy and the difference in predictions appear relatively small. This can be even more concerning, as students with similar competency could end up with very different interventions based on such predictions, further reinforcing existing educational inequalities. These findings also reinforce the long-recognized importance of considering the multi-faceted nature of predictive bias in education. By using fairness metrics that account for full distributional differences rather than just aggregate statistics, educators and policymakers can better identify and address inequities that might otherwise go unnoticed. This has important implications for educational interventions, as it highlights the need for predictive models that are not only accurate on average but also equitable across all student groups, particularly those at the margins of academic performance.

Beyond the methodological contributions of the proposed metrics, their practical value is also significant for educational stakeholders. For educators, it has long been important but also difficult to monitor whether automated predictive systems (such as essay scoring or GPA

prediction models) systematically overestimate or underestimate students’ performance Gulson et al. (2022). This difficulty mainly arises because existing evaluation metrics do not offer an intuitive and comprehensive way to understand fairness differences between student subgroups. The proposed metrics, *AreaCDF* and *AreaPDF*, address this gap by quantifying disparities across the full distribution of predicted outcomes rather than focusing solely on average performance, coupled with the long-established, easy-to-interpret visualization formats of CDF and density plots. This combination enables educators to clearly observe where disparities occur, whether prediction errors are concentrated among lower- or higher-performing students—and to implement more targeted and equitable instructional interventions accordingly. Moreover, the flexibility in defining groupings offered by conditional metrics such as *AreaPDF*-CSA and *AreaPDF*-CSD provides greater practical value for educational applications. In practice, this flexibility gives educators the freedom to define grouping schemes that align with their specific instructional needs and classroom contexts. For example, by organizing students according to grading bands, achievement tiers, or subject-specific performance ranges, thereby enabling fairness evaluations that are both interpretable and pedagogically meaningful. For policymakers and institutional leaders, these metrics can serve as diagnostic tools for auditing predictive systems used for student-related decision-making, ensuring that decisions informed by such systems do not inadvertently reinforce existing inequities. In large-scale assessment or resource-allocation contexts, these metrics can inform the design of evaluation frameworks that help ensure predictive systems are used responsibly, emphasizing both accuracy and fairness across all levels of student performance.

Results of debiasing methods also offer several key implications for fairness enhancements in continuous predictive modeling. The first key finding is that, contrary to expectations based on prior studies in classification debiasing, global demographic imbalance may not be the primary driver of predictive bias in continuous predictions. Moreover, simply achieving demographic balance in the training data is not a sufficient condition for ensuring more equitable outcomes. This can be supported by results from Tables 3, D.1–D.3. For example, despite the gender attribute having a higher level of balance in group representation than the disability attribute, both *AreaPDF*-OSA and *AreaPDF*-CSD exhibit a higher bias over gender than that over disability for numeracy predictions. Even after demographic balancing via DemoRW, these biases remain evident. This highlights the need for fairness interventions that go beyond demographic balancing and consider the interplay between demographic effects and underlying score distributions to mitigate biases effectively. The proposed reweighting-based data balancing methods directly address this challenge by considering both demographic group representation and score distributions in the balancing process. As demonstrated in Tables D.1–D.3, the approaches proposed in this study are generally effective across most fairness metrics and different Task-Attribute scenarios, reinforcing the necessity of integrating both demographic group representation and score distribution adjustments for effective fairness enhancement. Moreover, as indicated in Table 4, the proposed reweighting-based balancing methods do not exhibit a strict trade-off between predictive accuracy and fairness in most cases. In other words, these methods can enhance fairness without significantly compromising model performance in accuracy.

In addition to their overall effectiveness, the notably stronger performance of certain methods for specific demographic attributes and predictive models highlights the importance of selecting debiasing strategies that are well aligned with the intrinsic characteristics of both the data and the modeling approach. When demographic imbalances are more severe, more complex models such as MLP and XGBoost tend to achieve higher fairness, and the effects of the proposed debiasing methods become more pronounced. This pattern is evidenced by the lower bias observed in MLP and XGBoost predictions for disability-related scenarios and by the overall effectiveness of CondDemoBa1 in reducing bias across these scenarios. This differing impact of reweighting across

predictive models may be partly explained by how each model structures and combines features to make predictions. In real educational contexts, the effects of disability status on students' formative assessment outcomes are likely to be highly heterogeneous. For example, disability may interact differently with specific skills such as vocabulary recognition and counting, leading to distinct feature interactions associated with the demographic attribute. Tree-based models such as XGBoost and MLPs can capture nonlinear and interaction effects, reweighting changes the relative influence of samples in specific regions of the score distribution, which in turn leads to more substantial fairness gains. In contrast, linear models primarily adjust global feature weights, which may be less effective when bias is driven by local or interaction-based patterns. At the same time, caution should be exercised regarding the influence of feature composition on both accuracy and fairness when applying debiasing methods to the predictions of more complex models, as their higher capacity for identifying non-linear interactions may also amplify existing biases embedded in unbalanced or feature-limited data. This is evidenced by relatively higher accuracy loss and less effective fairness improvements of debiasing methods applied to MLP and XGBoost for the Numeracy-Gender scenario. Setting aside the effects of model architecture, a generally effective strategy for debiasing method selection emerging from our results is to prioritize CondDemoBal in scenarios with relatively balanced demographic distributions, and Demo-JointBal in scenarios characterized by substantial demographic imbalance.

Additionally, the importance of considering the distribution of task scores and demographic groups globally and locally should be emphasized especially when demographic subgroups are globally balanced such as gender in this study. By revisiting the three most effective methods for gender-related scenarios—CondDemoBal, Task-JointBal, and DTIndBal, further insight into the joint influence of task score distributions and demographic attributes becomes evident. These methods all share the common weighting term $\frac{p(y)}{p(y,g)}$, which primarily targets local demographic imbalances within each task score. Notably, when group representation is balanced, as in the case of gender in this study ($p(g = \text{Female}) \approx p(g = \text{Male})$), the demographic factor $p(g)$ in DTIndBal becomes negligible, effectively reducing it to Task-JointBal. The superior performance of CondDemoBal in such scenarios likely stems from its additional incorporation of global task score frequencies, capturing disparities arising from uneven score distributions even when demographic representation is balanced. This underscores the importance of considering both global task score distributions and their local associations with demographic groups when designing debiasing strategies for educational prediction tasks.

In summary, based on these findings, several recommendations for tackling predictive bias in continuous predictions include: (i) bias in continuous predictions should be measured beyond group averages to consider distributional disparities across the full range of predicted outcomes, especially for educational settings where students performances may significantly vary, (ii) it is particularly important to consider the interplay between demographic effects and score distribution in data balancing for effective debiasing, and (iii) more importantly, effective fairness interventions require context-sensitive strategies, tailored to the specific patterns of demographic representation and score distribution in the dataset and predictive models. When demographic group representation is globally balanced but score distribution is non-uniform, it is important to account for both the global frequency of scores and the local demographic representation within each score category to achieve fairness improvements. Conversely, when both demographic group representation and score distribution are uneven, it's worth considering both global demographic representation and its joint distribution with task scores to enhance fairness improvements. In other words, no single debiasing method is universally optimal; rather, the choice of strategy should consider factors such as feature composition, sample balance in terms of both task scores and demographic attributes, and model complexity to ensure fairness improvements without compromising predictive accuracy.

9. Conclusion and limitations

This study explored the critical yet often overlooked issue of fairness in continuous predicted scores, particularly in the context of predicting young children's national standardized test scores. The analysis specifically focused on biases related to gender (male vs. female) and disability status (with disabilities vs. without disabilities) in the prediction of students' early numeracy and literacy competencies. To overcome the limitations of existing metrics that rely on summary statistics and fail to capture the full extent of disparities between two distributions, this study proposed two novel metrics: *AreaCDF* and *AreaPDF*. These metrics assess bias across the entire distribution from cumulative distribution and probability density perspectives, respectively. Their broad applicability to any two continuous distributions enables a comprehensive analysis of bias across four predicted outcome dimensions: *overall scoring accuracy*, *overall scoring difference*, *conditional scoring accuracy*, and *conditional scoring difference*. Additionally, this study developed five reweighting-based data balancing methods to account for various distributional characteristics related to both demographic attributes and task scores, demonstrating their effectiveness in mitigating predictive bias.

This study also acknowledges certain limitations. First, due to their distance-based formulation, the proposed metrics are currently applied and discussed only in two-group settings. Extending these approaches to multi-group or intersectional scenarios introduces additional methodological and computational challenges, and remains an important direction for future work. Secondly, this study specifically focused on distribution-related biases; however, other forms of data bias, such as hardness bias (Yu et al., 2021) or label bias (Jiang & Nachum, 2020), can also contribute to unfairness. Third, the study examined only three predictive models and one dataset, which may limit the generalizability of the findings. Therefore, future efforts are expected to explore methods to extend these metrics for multi-group comparisons and investigate additional sources of bias that may impact fairness in educational predictions. In addition, future research would further examine the generalizability of the proposed metrics and debiasing approaches across diverse datasets, educational contexts, and model architectures to ensure their broader applicability. Another interesting direction would be to examine the effects of different approaches to estimating sample probabilities in the weighting process; in this study, probabilities were derived based on group-level counts, but alternative formulations (e.g., density- or model-based estimates) may yield different fairness behaviors.

CRedit authorship contribution statement

Lin Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Namrata Srivastava:** Writing – review & editing, Supervision. **Jia Rong:** Writing – review & editing, Supervision. **Quanlong Guan:** Writing – review & editing. **Dragan Gašević:** Writing – review & editing, Supervision. **Guanliang Chen:** Writing – review & editing, Supervision.

Statements on open data and ethics

The study was approved by an ethics committee with ID: 37433. Informed consent was obtained from all participants, and their privacy rights were strictly observed.

Ethics approval statement

The data we used has passed the critical ethical inspection by the University ethics committee and obtained ethical approval.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Dataset and model

In this section, we provide more details regarding the feature engineering and feature description.

A.1. Feature engineering

In prior studies, parents' education and occupation were demonstrated to affect students' early skill development (Thomson, 2018), and school-level performance such as average test performance was also examined to have an effect (Incognito & Pinto, 2023). Therefore, in this study we engineered three groups of features related to: (1) Formative Assessment performance, (2) Family Background, and (3) School Characteristics. All the features are listed in Table A.1. In terms of features related to *Formative Assessment performance*, 52 numerical features corresponding to the scores on 12 formative assessments across Kindergarten, Year 1, and Year 2 were included. For *Family-related features* we counted the number of parents with an education level above Year 9, the number of parents with a qualification of diploma or above, and the number of parents whose occupations require professional skills. Besides, we also included the number of siblings and sibling order as informed by previous studies. Regarding *School-related features*, we considered the school average performance on the five categories of the NAPLAN test in the last year, covering Numeracy, Reading, Writing, Grammar, and Spelling, as well as the SES of the school. Detailed descriptions of the engineered features are provided in Section 1.2 below. Note that when building the linear regression model, all numerical features were rescaled using min-max scaling, given their varied magnitudes (Mining, 2006). Similarly, the prediction target variables, i.e., numeracy and reading test scores, were also rescaled to 0 ~ 1 to ensure better numerical stability considering features had been rescaled to 0 ~ 1.

Table A.1
Features engineered for predictive modeling.

Category	Feature	Type	Description	Scale
Assessments	Addition&Subtraction	Integer	Quantify students ability to solve addition and subtraction problems	0~6
	Counting	Integer	Quantify students understanding of the number sequence	0~6
	Multiplication& Division	Integer	Quantify students ability to solve multiplication and division problems	0~6
	Place Value	Integer	Quantify students understanding of each digit in numbers of various size	0~6
	Accuracy	Decimal	Quantify the errors a student made when reading a given text	0~1
	SelfCorrectionRate	Decimal	Quantify the self corrections a student made when reading a given text	
	TextLevel	Integer	The difficulty level of the given text used to test students' reading	0~31
	Concepts About Print (CAP)	Integer	Consist of 24 items to assess students reading readiness skills and concepts, e.g., finding the front of the book, knowing the difference between a picture and text	0~24
	Letter identification	Integer	Assess what letters student knows and how they can identify, both capital and lowercase	0~54
	Burt Word Test	Integer	Test the number of words a student can recognize	0~110
	Clay Word Test	Integer	15 words are selected for students to identify	0~15
	HRSIW	Integer	Assess phonemic awareness by determining if the child can hear individual sounds buried within words and represents sounds with appropriate written symbols	0~50
	WritingVocab	Integer	Asks the student to write down all of the words he or she knows	No Maximum
	PAT-R	Integer	A test of reading comprehension and word knowledge	
Family	#Parents with EduLevel > Y9	Integer	Number of parents with educational level above Year 9	
	#Parents with QauLevel > Diploma	Integer	Number of parents with qualification level above Diploma	
	#Parents with Occ on professional	Integer	Number of parents with professional-level occupation	
	#Sibling	Integer	Number of siblings	
	SiblingOrder	Integer	Rank among the kids in the family	
School	Last-year school mean NAPLAN-numeracy	Decimal	NAPLAN mean score of the school last year in numeracy	0~1000
	Last-year school mean NAPLAN-reading	Decimal	NAPLAN mean score of the school last year in reading	0~1000
	Last-year school mean NAPLAN-writing	Decimal	NAPLAN mean score of the school last year in writing	0~1000
	SES/CTC	Integer	A score to measure the the school community's capacity to contribute financially to the school	

A.2. Feature description

In this subsection, we provide a detailed description of each category of features including their background and original forms.

A.2.1. Student performance on formative assessments

This study includes student performance over a set of formative assessments that are either numeracy-oriented or literacy-oriented.

Numeracy-oriented assessments: a clinical interview named Mathematics Assessment Interview (MAI) is normally conducted by teachers at the beginning of a school year in a one-to-one interview scenario to determine students mathematical knowledge in four number domains including Addition & Subtraction, Counting, Multiplication, and Place Value, with each detailed below.

- Addition & Subtraction assesses student mastery of basic addition and subtraction facts where students might be asked questions involving single-digit or double-digit addition and subtraction, or to explain how they approach the problems to understand their thought processes;
- Counting measures student's understanding of number relationships and sequences, and their ability to apply counting principles to various mathematical situations by asking students to count objects or count back/forth
- Multiplication & Division measures students' ability to solve multiplication and division problems and their understanding of these concepts;
- Place Value measures students comprehension and understanding of multi-digit numbers, e.g., they may be asked to point out the position of the digit representing the tens, hundreds or thousands in a given number. Each student will be assigned a score scaled from 0 to 6 on these four assessments according to their responses.

Literacy-oriented assessments, the Early Years Assessment (EYA) guided by An Observation Survey of Early Literacy Achievement are used to assess students knowledge of early reading and writing in the form of teacher observation of children's responses in six literacy tasks, including the following.

- Text Reading works by asking students to read a text of a specific difficulty level (scaled 0 to 31) that is selected by teachers and appropriate for the student reading level, which is scheduled three times during each school year, i.e., the start of the year (SOY), the middle of the year (MOY), and the end of the year (EOY), and can be analysed to assess student reading proficiency according to the Accuracy, i.e., the proportion of correctly read words, and Self Correction Rate.
- Concepts About Print (CAP) determines what a child understands about how written language works such as the direction of reading printed materials, how to turn pages and follow the text on a page, and the recognition of letters and words.
- Letter Identification (LetterID) determines which letters a child knows and how to identify them by name, sound or within a word. Both CAP and LetterID are scheduled two times per year, i.e., at SOY and EOY.
- Clay Word Reading (Clay) determines students' understanding of word structure and their ability to analyse and manipulate the phonological and orthographic aspects of words;
- Hearing and Recording Sounds in Words (HRSIW) assesses phonemic awareness and phonic skills by testing if a child can spell or write down the words they hear.
- Writing Vocabulary (WritingVocab) determines what words a child can write.

In addition to these six observation tasks, another two reading-oriented assessments, namely Burt Word Test and Progressive Achievement Tests in Reading (PAT-R), are also used,

- Burt Word Test measures a child's vocabulary size.
- PAT-R is a more standardized reading assessment administered by teachers in a classroom to assess students' reading comprehension abilities, which requires students to read the passage and answer a few questions related to the text.

Note that Clay, HRSIW, Burt and WritingVocab are scheduled two times only in Year 1, i.e., SOY and EOY, while PAT-R is conducted in Year 2.

A.2.2.2. Family background

Parents' education level, qualification level, and occupation level are also collected for each student. Specifically, six different levels are recorded for education including "Junior Certificate," "Year 9 or equivalent or below," "Year 10 or equivalent," "Year 11 or equivalent," "Year 12 or equivalent," and "Higher Certificate." Qualification has four levels, namely "No Non-school Qualification" (meaning not holding any formal certifications or qualifications), "Certificate I to IV (Including Trade Certificate)," "Advanced Diploma/Diploma," and "Bachelor Degree or Above." Occupation also has four different categories including "Machine Operators, hospitality staff, assistants, labourers and related workers," "Tradesmen/women, clerks and skilled office, sales and service staff," "Other business managers, arts/media/sportspersons and associate professionals," and "Senior management in large business organisations, government administration and defence and qualified professionals." In addition, the number of siblings and sibling order are also recorded for each student.

A.2.3. School characteristics

For each school, we have its overall performance on past NAPLAN tests, including scores in numeracy, reading, writing, spelling, and grammar. Additionally, each school is associated with a value representing the socio-economic status (SES) of the school. Note that the SES value is now referred to as Capacity to Contribute, which assesses the financial capacity of families to contribute to the cost of their child's education in a non-government (independent or Catholic) school and is considered when determining the level of government funding.

Appendix B. Analysis of existing metrics

B.1. Description and calculation of R^2 -based fairness metrics

To evaluate bias in machine-generated scores on students' essays compared to human scores, previous studies (Litman et al., 2021; Loukina et al., 2019) defined bias as the demographic effect on the outcome variable (e.g., score accuracy or score difference), quantified by the R^2 from a simple linear regression model where the outcome is regressed on demographic group membership. Specifically, when the fairness focus is on *overall score accuracy*, i.e., whether predicted scores are equally accurate for students of different demographic groups, the bias is the R^2 of the linear regression model fitting squared errors using demographic group membership. For *overall score difference*, i.e., whether predicted scores are consistently different from human scores for students of different demographic groups, the bias is the R^2 of the linear regression model fitting signed errors using demographic group membership. In terms of *conditional score difference*, the bias is the R^2 difference between two linear regression models, where one only uses actual scores to fit score difference, and another uses both actual scores and demographic group membership as predictors.

B.2. Limitation of R^2 -based fairness metrics

One primary limitation of these R^2 -based metrics is that when the demographic attribute is binary (as is common in most fairness studies), they reduce to a value that merely captures the group-averaged difference in the outcome variable, failing to account for distributional differences between groups. We demonstrate this limitation through the mathematical proof provided below.

Theorem 2. Consider a simple linear regression model with the form $\hat{y} = w_0 + w_1x$, where $\hat{y}$ is the predicted outcome, x is a binary independent variable indicating demographic group membership, (e.g., $x \in \{0, 1\}$ for male vs. female), w_0 is the intercept, and w_1 is the regression coefficient. Let y be the true outcome values. Then, the coefficient of determination R^2 of this model is a function of the difference between the group-wise average values of the actual scores y , and does not capture distributional differences beyond the means.

Proof. According to the ordinary least squares (OLS) regression, the weight w_1 can be computed as:

$$w_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad (\text{B.1})$$

where i represents the sample index. When x corresponds to the binary group membership with values 0 or 1, let the total size of the sample be n , the size of samples in the group $G_0(x = 0)$ be n_0 , and the size of group $G_1(x = 1)$ be n_1 . In this case, the mean of x i.e., $\bar{x}$, represents the proportion of samples in group G_1 , i.e., $\frac{n_1}{n}$ or $\frac{n_1}{n_0+n_1}$. Hence, the denominator of w_1 , i.e., $\sum_i (x_i - \bar{x})^2$ can be re-written as follows:

$$\sum_i (x_i - \bar{x})^2 = \sum_{i \in G_1} (x_i - \bar{x})^2 + \sum_{i \in G_0} (x_i - \bar{x})^2 \quad (\text{B.2})$$

$$= \sum_{i \in G_1} \left(1 - \frac{n_1}{n_1 + n_0}\right)^2 + \sum_{i \in G_0} \left(0 - \frac{n_1}{n_1 + n_0}\right)^2 \quad (\text{B.3})$$

$$= \sum_{i \in G_1} \left(\frac{n_0}{n_1 + n_0}\right)^2 + \sum_{i \in G_0} \left(\frac{n_1}{n_1 + n_0}\right)^2 \quad (\text{B.4})$$

$$= n_1 \cdot \left(\frac{n_0}{n_1 + n_0}\right)^2 + n_0 \cdot \left(\frac{n_1}{n_1 + n_0}\right)^2 \quad (\text{B.5})$$

$$= \frac{n_1 \cdot n_0(n_0 + n_1)}{(n_1 + n_0)^2} \quad (\text{B.6})$$

$$= \frac{n_1 \cdot n_0}{n_1 + n_0}. \quad (\text{B.7})$$

Correspondingly, the numerator of w_1 , i.e., $\sum_i (x_i - \bar{x})(y_i - \bar{y})$, can be rewritten as follow:

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i \in G_1} (x_i - \bar{x})(y_i - \bar{y}) + \sum_{i \in G_0} (x_i - \bar{x})(y_i - \bar{y}) \quad (\text{B.8})$$

$$= \sum_{i \in G_1} \left(1 - \frac{n_1}{n_1 + n_0}\right) (y_i - \bar{y}) + \sum_{i \in G_0} \left(0 - \frac{n_1}{n_1 + n_0}\right) (y_i - \bar{y}) \quad (\text{B.9})$$

$$= \sum_{i \in G_1} \frac{n_0}{n_1 + n_0} (y_i - \bar{y}) + \sum_{i \in G_0} \left(-\frac{n_1}{n_1 + n_0}\right) (y_i - \bar{y}) \quad (\text{B.10})$$

$$= \frac{n_0}{n_1 + n_0} \sum_{i \in G_1} (y_i - \bar{y}) - \frac{n_1}{n_1 + n_0} \sum_{i \in G_0} (y_i - \bar{y}) \quad (\text{B.11})$$

$$= \frac{n_0}{n_1 + n_0} \left(\sum_{i \in G_1} y_i - \sum_{i \in G_1} \bar{y} \right) - \frac{n_1}{n_1 + n_0} \left(\sum_{i \in G_0} y_i - \sum_{i \in G_0} \bar{y} \right) \quad (\text{B.12})$$

$$= \frac{n_0}{n_1 + n_0} (n_1 \bar{y}_1 - n_1 \bar{y}) - \frac{n_1}{n_1 + n_0} (n_0 \bar{y}_0 - n_0 \bar{y}) \quad (\text{B.13})$$

$$= \frac{n_0 \cdot n_1}{n_1 + n_0} \bar{y}_1 - \frac{n_0 \cdot n_1}{n_1 + n_0} \bar{y}_0 - \frac{n_0 \cdot n_1}{n_1 + n_0} \bar{y} + \frac{n_0 \cdot n_1}{n_1 + n_0} \bar{y} \quad (\text{B.14})$$

$$= \frac{n_0 \cdot n_1}{n_1 + n_0} (\bar{y}_1 - \bar{y}_0) \quad (\text{B.15})$$

Substituting above two to Eq. (B.1), w_1 essentially reduces to $w_1 = \bar{y}_1 - \bar{y}_0$. Based on this and the fact that the linear regression line must pass through the point $(\bar{x}, \bar{y})$, w_0 can be further derived as

$$\bar{y} = w_0 + w_1 \bar{x} \quad (\text{B.16})$$

$$\Rightarrow w_0 = \bar{y} - (\bar{y}_1 - \bar{y}_0) \frac{n_1}{n_1 + n_0} = \frac{n_1 \bar{y}_1 + n_0 \bar{y}_0}{n_1 + n_0} - (\bar{y}_1 - \bar{y}_0) \frac{n_1}{n_1 + n_0} \quad (\text{B.17})$$

$$\Rightarrow w_0 = \bar{y}_0 \quad (\text{B.18})$$

Substituting w_0 and w_1 into the Eq. (B.1), the predicted score $\hat{y}$ can be calculated as:

$$\hat{y} = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x \quad (\text{B.19})$$

$$\text{when } x = 1, \hat{y} = \bar{y}_1, \text{ and when } x = 0, \hat{y} = \bar{y}_0 \quad (\text{B.20})$$

Based on this and $R^2 = 1 - \frac{RSS}{TSS}$, where $RSS = \sum_i (\hat{y}_i - y_i)^2$ and $TSS = \sum_i (y_i - \bar{y})^2$, i.e., only RSS is related to predicted score, which can be further rewritten as:

$$RSS = \sum_i (\hat{y}_i - y_i)^2 = \sum_{i \in G_1} (\hat{y}_i - y_i)^2 + \sum_{i \in G_0} (\hat{y}_i - y_i)^2 \quad (\text{B.21})$$

$$= \sum_{i \in G_1} (\hat{y}_i^2 - 2\hat{y}_i y_i + y_i^2) + \sum_{i \in G_0} (\hat{y}_i^2 - 2\hat{y}_i y_i + y_i^2) \quad (\text{B.22})$$

$$= \sum_{i \in G_1} \hat{y}_i^2 - \sum_{i \in G_1} 2\hat{y}_i y_i + \sum_{i \in G_1} y_i^2 + \sum_{i \in G_0} \hat{y}_i^2 - \sum_{i \in G_0} 2\hat{y}_i y_i + \sum_{i \in G_0} y_i^2 \quad (\text{B.23})$$

$$= \sum_{i \in G_1} \bar{y}_1^2 - \sum_{i \in G_1} 2\bar{y}_1 y_i + \sum_{i \in G_1} y_i^2 + \sum_{i \in G_0} \bar{y}_0^2 - \sum_{i \in G_0} 2\bar{y}_0 y_i + \sum_{i \in G_0} y_i^2 \quad (\text{B.24})$$

$$= n_1 \bar{y}_1^2 - 2n_1 \bar{y}_1 + \sum_{i \in G_1} y_i^2 + n_0 \bar{y}_0^2 - 2n_0 \bar{y}_0 + \sum_{i \in G_0} y_i^2 \quad (\text{B.25})$$

$$= \sum_i y_i^2 - (n_1 \bar{y}_1^2 + n_0 \bar{y}_0^2) \quad (\text{B.26})$$

$$= \sum_i y_i^2 - (n_1 \bar{y}_1^2 + (n - n_1) \bar{y}_0^2) \quad (\text{B.27})$$

$$= \sum_i y_i^2 - [n_1 (\bar{y}_1^2 - \bar{y}_0^2) + n \cdot \bar{y}_0^2] \quad (\text{B.28})$$

Note that the denominator i.e., $TSS = \sum_i (y_i - \bar{y})^2$ only involves the actual scores y , which can be considered a constant, thus the only changeable part related to demographic group membership is the RSS, which, as seen above, is essentially a combination of the group-averaged actual scores, including $\bar{y}_1$ and $\bar{y}_0$. $\square$

Therefore, regardless of whether the outcome variable y to be examined is score accuracy (i.e., squared or absolute error) or score difference (i.e., signed prediction error), R^2 -based metrics essentially capture differences in group-averaged outcome values, failing to account for distributional differences between groups.

Appendix C. Implementation details

R^2 -based Metrics. Note that, in the original paper where R^2 -based metrics were first proposed (Litman et al., 2021), R^2 -OSA was calculated by fitting squared prediction errors based on group membership. In this study, for OSA, we adopted absolute errors instead of squared errors to facilitate a direct comparison and to better reflect the actual magnitude of prediction errors without exaggerating the impact of larger errors. For all R^2 -based metrics, i.e., R^2 -OSA, R^2 -OSD, and R^2 -CSD, we implemented the same computation process in RStudio following the original implementation written in Python provided in <https://github.com/EducationalTestingService/rsmtool>.

AreaCDF & AreaPDF. For *AreaCDF*, we followed a similar approach used in `stats::ecdf()` from the `ggplot2` package, which computes the empirical CDF by sorting the unique observations, accumulating their frequencies, and normalizing by the sample size to obtain the cumulative probability. We then apply `stats::approxfun()` to obtain a continuous, linearly interpolated approximation of the (stepwise) empirical CDF for integration and evaluation. For *AreaPDF*, we estimate densities using kernel density estimation (e.g., `stats::density()` with the default kernel function (Gaussian) and bandwidth selectors (Silverman's rule). For conditional metrics, we excluded bins where any group has fewer than 3 samples from the conditional metric calculation. This choice prevents high-variance estimates from disproportionately influencing the aggregate metric. For *AreaPDF-CSA* and *AreaPDF-CSD*, we first group students into ten bins based on equal ranges derived from the minimum and maximum predicted outcomes, i.e., dividing the range of predicted values into ten intervals of equal width. Note that this grouping can also be adapted based on domain expertise. For each performance subgroup, we calculated *AreaPDF-OSA* (or *AreaPDF-OSD*), and averaged these values as the final result of *AreaPDF-CSA* (or *AreaPDF-CSD*).

For the probabilities used in the proposed re-weighting approaches, we adopted a straightforward method, estimating probabilities by directly calculating the ratio of occurrences of exact values within the dataset.

Appendix D. Supplementary tables and figures for results

We also provided visualizations of *AreaCDF* and *AreaPDF* calculated over actual scores in the training data across the four task-attribute scenarios, i.e., numeracy and reading test scores between students of different genders (male vs. female) and between students of different disability statuses (with disabilities vs. without) in Fig. D.1, from which it can be observed that, compared to gender, students with different disability statuses tend to have more pronounced differences in overall numeracy and reading competency. Specifically, students with disabilities generally exhibit lower performance in both numeracy and reading tests.

In Fig. D.2 we visualized *AreaPDF-OSD* across the four task-attribute scenarios to get a finer-grained view of how disparities are distributed across the whole range of score difference distributions. It can be observed that, for numeracy test score predictions, while female students are more likely than male students to be over- or under-estimated within

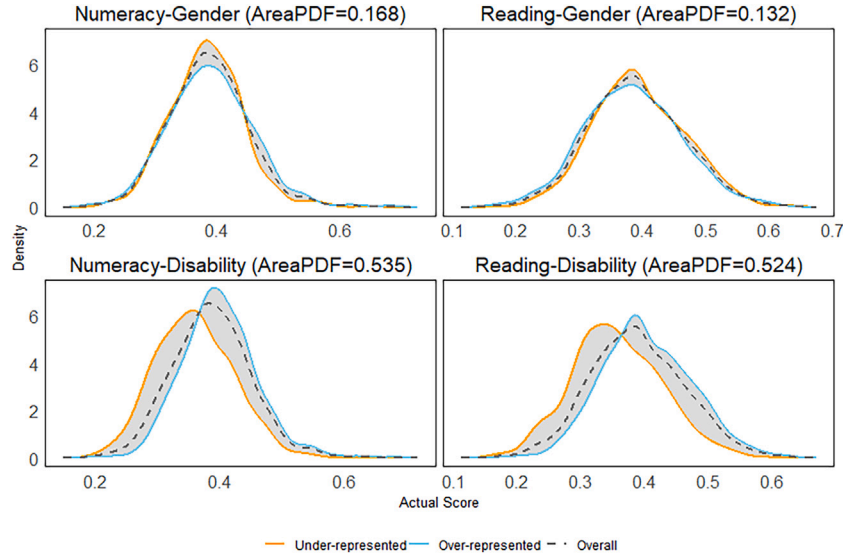


Fig. D.1. Illustration of *AreaPDF* calculated on the actual scores in the training data for the four Task-Attribute scenarios, i.e., numeracy-gender, reading-gender, numeracy-disability, and reading-disability, where the gray shaded area represents how student performance in numeracy or reading tests is distributed differently across different groups. The under-represented group for gender corresponds to females, and for disability corresponds to students with disabilities. The dashed line represents the overall distribution of student test scores without grouping. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

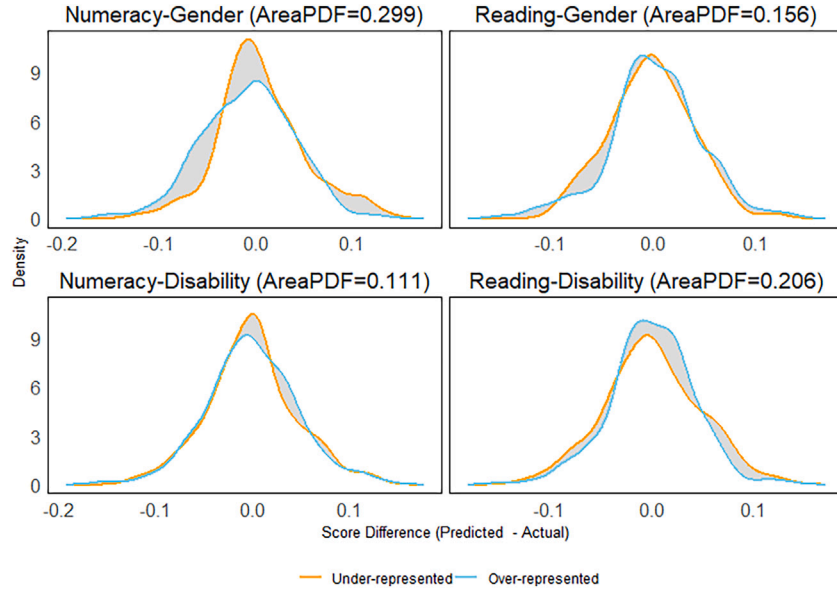


Fig. D.2. Illustration of **bias in overall score difference** measured by *AreaPDF-OSA* (i.e., gray shaded area) across the four Task-Attribute scenarios, i.e., numeracy-gender, reading-gender, numeracy-disability, and reading-disability. Values over x axis are prediction errors, i.e., predicted scores - actual scores. Under-represented group for gender corresponds to females, and for disability corresponds to students with disabilities. The dashed line represents the distribution of all test scores.

smaller score differences, such as around 30 marks, they also have a higher likelihood of being over-estimated by a larger margin, such as 100 marks (i.e., 0.1 in the x-axis). Additionally, these subplots highlight

that the bias related to gender attributes on the OSA of numeracy predictions is the largest among the four scenarios, despite male and female students being similarly represented in the training data.

Predictive fairness measured by the proposed metrics over predictions from LR, with each row showing results across all debiasing methods for a given metric examined in a task-attribute scenario. The last four rows (Rows 25–28) report the number of cases (out of 24) in which each corresponding debiasing method (shown in columns) achieved the best fairness without and with statistical significance, as well as the number of cases showing fairness improvement without and with statistical significance, respectively. The `Value` column presents the point estimate of bias measured by the corresponding metric, with values in parentheses indicating the 95% confidence interval. The `Bias-reduction%` column reports the percentage reduction in bias relative to the unbiased setting (with 95% confidence intervals in parentheses), where positive values indicate improved fairness (i.e., reduced bias). Cases of reduced bias are highlighted in green. The `#` symbol denotes the greatest fairness improvement among all debiasing methods. * indicates that the point estimate lies within the 95% confidence interval, and the interval does not include 0, suggesting that the observed fairness improvement is statistically significant and consistent with the estimated effect direction.

Task-Attribute	RowID	Metric	Unlabeled		Previous Metrics										Proposed Metrics									
			LR	DeltaMWR	Value (95% CI)	Reichleids	Value (95% CI)	Flaring	Value (95% CI)	Restoration	Value (95% CI)	Journal	Task-Journal	Value (95% CI)	Delta-Journal	Value (95% CI)	Confidentiality	OffThreats						
Numeracy-Gender	1	Assoc-CFSA	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)	0.006 (0.000-0.008)						
	2	Assoc-CFSA	0.275 (0.014-0.385)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)	0.275 (0.014-0.387)						
	3	Assoc-CFSA	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)						
	4	Assoc-CFSA	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)	0.004 (0.000-0.008)						
	5	Assoc-CFSA	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)	0.534 (0.36-0.580)						
Reading-Gender	7	Assoc-CFSA	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)						
	8	Assoc-CFSA	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)	0.103 (0.007-0.203)						
	9	Assoc-CFSA	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)						
	10	Assoc-CFSA	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)	0.156 (0.09-0.208)						
	11	Assoc-CFSA	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)	0.032 (0.008-0.058)						
Numeracy-Disability	13	Assoc-CFSA	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)	0.003 (0.002-0.004)						
	14	Assoc-CFSA	0.138 (0.008-0.189)	0.138 (0.007-0.182)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)	0.138 (0.008-0.189)						
	15	Assoc-CFSA	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)	0.007 (0.001-0.011)						
	16	Assoc-CFSA	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)	0.111 (0.051-0.161)						
	18	Assoc-CFSA	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)	0.010 (0.003-0.017)						
Reading-Disability	24	Assoc-CFSA	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)	0.030 (0.04-0.777)						
	25	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	26	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	27	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	28	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
Unlabeled	29	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	30	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	31	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	32	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						
	33	Assoc-CFSA	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)	0.000 (0.000-0.000)						

Predictive fairness measured by the proposed metrics over predictions from **MLP**, with each row showing results across all debiasing methods for a given metric examined in a task-attribute scenario. The last four rows (Rows 25–28) report the number of cases (out of 24) in which each corresponding debiasing method (shown in columns) achieved the best fairness without and with statistical significance, as well as the number of cases showing fairness improvement without and with statistical significance, respectively. The **Value** column presents the point estimate of bias measured by the corresponding metric, with values in parentheses indicating the 95% confidence interval. The **Bias-reduction%** column reports the percentage reduction in bias relative to the unbiased setting (with 95% confidence intervals in parentheses), where positive values indicate improved fairness (i.e., reduced bias). Cases of reduced bias are highlighted in green. The $\$$ symbol denotes the greatest fairness improvement among all debiasing methods. * indicates that the point estimate lies within the 95% confidence interval, and the interval does not include 0, suggesting that the observed fairness improvement is statistically significant and consistent with the estimated effect direction.

[illegible]

Table D.3

Predictive fairness measured by the proposed metrics over predictions from **XGBOOST**, with each row showing results across all debiasing methods for a given metric examined in a task-attribute scenario. The last four rows (Rows 25–28) report the number of cases (out of 24) in which each corresponding debiasing method (shown in columns) achieved the best fairness without and with statistical significance, as well as the number of cases showing fairness improvement without and with statistical significance, respectively. The *value* column presents the point estimate of bias measured by the correspondence metric, with values in parentheses indicating the 95% confidence interval. The *Bias-reduction%* column reports the percentage reduction in bias relative to the unbiased setting (with 95% confidence intervals in parentheses), where positive values indicate improved fairness (i.e., reduced bias). Cases of reduced bias are highlighted in green. The *symbol* denotes the greatest fairness improvement among all debiasing methods. * indicates that the point estimate lies within the 95% confidence interval, and the interval does not include 0, suggesting that the observed fairness improvement is statistically significant and consistent with the estimated effect direction.

Item-ID	Risk Attribute	Previous Methods										Proposed Method			Confidence			FT/Fall	
		DenseRW		Radar		Interf		Task-Interf		Bene-Interf		Interf-Interf		Bene-Interf		Confidence			
Metrics	XGB-OST	Unlabeled	Value (95% CI)		Value (95% CI)		Value (95% CI)		Value (95% CI)		Value (95% CI)		Value (95% CI)		Value (95% CI)		Value (95% CI)		
			Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)	Value (95% CI)		
			0.003 (0.002, 0.004)	0.001 (0.000, 0.002)	-33.9% (-328.9% - 17.6%)	-0.003 (0.002, 0.000)	-66.7% (-280.7% - 6.4%)	0.003 (0.000, 0.006)	-66.7% (-280.7% - 6.4%)	0.005 (0.000, 0.008)	-66.7% (-280.7% - 6.4%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)
			0.139 (0.083 - 0.183)	0.185 (0.083 - 0.287)	-33.1% (-134.6% - 1.9%)	0.212 (0.073 - 0.318)	-52.2% (-288.5% - 10.4%)	0.201 (0.083 - 0.310)	-43.9% (-242.5% - 14.4%)	0.176 (0.090 - 0.251)	-26.6% (-182.2% - 15.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)
			0.016 (0.008 - 0.025)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.016 (0.008 - 0.025)	-4.3% (-23.0% - 9.4%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)
			0.427 (0.315 - 0.491)	0.427 (0.315 - 0.491)	-4.2% (-23.2% - 15.3%)	0.446 (0.380 - 0.641)	-28.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)
Bene-Disability	XGB-OST	Unlabeled	0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
			0.003 (0.002, 0.004)	0.001 (0.000, 0.002)	-33.9% (-328.9% - 17.6%)	-0.003 (0.002, 0.000)	-66.7% (-280.7% - 6.4%)	0.003 (0.000, 0.006)	-66.7% (-280.7% - 6.4%)	0.005 (0.000, 0.008)	-66.7% (-280.7% - 6.4%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)
			0.139 (0.083 - 0.183)	0.185 (0.083 - 0.287)	-33.1% (-134.6% - 1.9%)	0.212 (0.073 - 0.318)	-52.2% (-288.5% - 10.4%)	0.201 (0.083 - 0.310)	-43.9% (-242.5% - 14.4%)	0.176 (0.090 - 0.251)	-26.6% (-182.2% - 15.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)
			0.016 (0.008 - 0.025)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.016 (0.008 - 0.025)	-4.3% (-23.0% - 9.4%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)
			0.427 (0.315 - 0.491)	0.427 (0.315 - 0.491)	-4.2% (-23.2% - 15.3%)	0.446 (0.380 - 0.641)	-28.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
Bene-Disability	XGB-OST	Unlabeled	0.003 (0.002, 0.004)	0.001 (0.000, 0.002)	-33.9% (-328.9% - 17.6%)	-0.003 (0.002, 0.000)	-66.7% (-280.7% - 6.4%)	0.003 (0.000, 0.006)	-66.7% (-280.7% - 6.4%)	0.005 (0.000, 0.008)	-66.7% (-280.7% - 6.4%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)
			0.139 (0.083 - 0.183)	0.185 (0.083 - 0.287)	-33.1% (-134.6% - 1.9%)	0.212 (0.073 - 0.318)	-52.2% (-288.5% - 10.4%)	0.201 (0.083 - 0.310)	-43.9% (-242.5% - 14.4%)	0.176 (0.090 - 0.251)	-26.6% (-182.2% - 15.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)
			0.016 (0.008 - 0.025)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.016 (0.008 - 0.025)	-4.3% (-23.0% - 9.4%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)
			0.427 (0.315 - 0.491)	0.427 (0.315 - 0.491)	-4.2% (-23.2% - 15.3%)	0.446 (0.380 - 0.641)	-28.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
Bene-Disability	XGB-OST	Unlabeled	0.003 (0.002, 0.004)	0.001 (0.000, 0.002)	-33.9% (-328.9% - 17.6%)	-0.003 (0.002, 0.000)	-66.7% (-280.7% - 6.4%)	0.003 (0.000, 0.006)	-66.7% (-280.7% - 6.4%)	0.005 (0.000, 0.008)	-66.7% (-280.7% - 6.4%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)
			0.139 (0.083 - 0.183)	0.185 (0.083 - 0.287)	-33.1% (-134.6% - 1.9%)	0.212 (0.073 - 0.318)	-52.2% (-288.5% - 10.4%)	0.201 (0.083 - 0.310)	-43.9% (-242.5% - 14.4%)	0.176 (0.090 - 0.251)	-26.6% (-182.2% - 15.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)
			0.016 (0.008 - 0.025)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.016 (0.008 - 0.025)	-4.3% (-23.0% - 9.4%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)
			0.427 (0.315 - 0.491)	0.427 (0.315 - 0.491)	-4.2% (-23.2% - 15.3%)	0.446 (0.380 - 0.641)	-28.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
Bene-Disability	XGB-OST	Unlabeled	0.003 (0.002, 0.004)	0.001 (0.000, 0.002)	-33.9% (-328.9% - 17.6%)	-0.003 (0.002, 0.000)	-66.7% (-280.7% - 6.4%)	0.003 (0.000, 0.006)	-66.7% (-280.7% - 6.4%)	0.005 (0.000, 0.008)	-66.7% (-280.7% - 6.4%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)
			0.139 (0.083 - 0.183)	0.185 (0.083 - 0.287)	-33.1% (-134.6% - 1.9%)	0.212 (0.073 - 0.318)	-52.2% (-288.5% - 10.4%)	0.201 (0.083 - 0.310)	-43.9% (-242.5% - 14.4%)	0.176 (0.090 - 0.251)	-26.6% (-182.2% - 15.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)
			0.016 (0.008 - 0.025)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.016 (0.008 - 0.025)	-4.3% (-23.0% - 9.4%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.022 (0.011 - 0.034)	-2.5% (-16.0% - 14.6%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)	0.015 (0.008 - 0.022)	1.4% (-8.4% - 11.2%)
			0.427 (0.315 - 0.491)	0.427 (0.315 - 0.491)	-4.2% (-23.2% - 15.3%)	0.446 (0.380 - 0.641)	-28.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
			0.428 (0.319 - 0.476)	0.445 (0.319 - 0.476)	-4.4% (-34.3 - 63.6)	-29.6% (-157.1% - 15.3%)	0.446 (0.385 - 0.491)	-4.3% (-30.5% - 30.8%)	0.412 (0.360 - 0.524)	-11.5% (-66.5% - 4.7%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.556 (0.316 - 0.677)	-29.2% (-213.2% - 4.9%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	0.564 (0.383 - 0.641)	25.3% (-102.0% - 15.5%)	
Bene-Disability	XGB-OST	Unlabeled	0.003 (0.002, 0.004)	0.001 (0.000, 0.002)	-33.9% (-328.9% - 17.6%)	-0.003 (0.002, 0.000)	-66.7% (-280.7% - 6.4%)	0.003 (0.000, 0.006)	-66.7% (-280.7% - 6.4%)	0.005 (0.000, 0.008)	-66.7% (-280.7% - 6.4%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.007 (0.003, 0.013)	-33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)	0.004 (0.002, 0.005)	33.9% (-328.9% - 17.6%)
			0.139 (0.083 - 0.183)	0.185 (0.083 - 0.287)	-33.1% (-134.6% - 1.9%)	0.212 (0.073 - 0.318)	-52.2% (-288.5% - 10.4%)	0.201 (0.083 - 0.310)	-43.9% (-242.5% - 14.4%)	0.176 (0.090 - 0.251)	-26.6% (-182.2% - 15.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.229 (0.093 - 0.364)	-47.7% (-260.5% - 4.5%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)	0.236 (0.109 - 0.360)	49.8% (-262.5% - 24.8%)
			0.016 (0.008 - 0.025)	0.017 (0.009 - 0.025)	-4.3% (-23.0% - 9.4%)	0.017 (0.009													

Data availability statement

We used one dataset for this study, which was collected from a large school system covering 58 primary schools, which dataset will not be publicly available due to regulations related to privacy and ethical guidelines. However, demo samples may be provided upon request.

References

Agarwal, A., Dudík, M. & Wu, Z. S. 2019. Fair regression: Quantitative definitions and reduction-based algorithms, in: *International Conference on Machine Learning*, PMLR. pp. 120–129.

Baker, R. S., & Hawn, A. 2021. Algorithmic bias in education. *International Journal of AIED*, 1–41.

Bakharia, A. 2016. Towards cross-domain mooc forum post classification, in: *Proceedings of the Third (2016) ACM Conference on Learning@ Scale*, pp. 253–256.

Barocas, S., Hardt, M., & Narayanan, A. 2023. *Fairness and machine learning: Limitations and opportunities*. MIT press.

Basu, A., Shioya, H., & Park, C. 2011. *Statistical inference: the minimum distance approach*. CRC press.

Berk, R., Heidari, H., Jabbari, S., Joseph, M., Kearns, M., Morgenstern, J., Neel, S., & Roth, A. 2017. A convex framework for fair regression. arXiv preprint arXiv:1706.02409.

Bird, S., Dudík, M., Edgar, R., Horn, B., Lutz, R., Milan, V., Sameki, M., Wallach, H., & Walker, K. 2020. Fairlearn: A toolkit for assessing and improving fairness in ai.

Cai, Y., & Lim, L.-H. 2022. Distances between probability distributions of different dimensions. *IEEE Transactions on Information Theory* 68, 4020–4031.

Caton, S., & Haas, C. 2020. Fairness in machine learning: A survey. arXiv preprint arXiv:2010.04053.

Chen, T., & Guestrin, C. 2016. Xgboost: A scalable tree boosting system, in: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794.

Chinta, S. V., Wang, Z., Yin, Z., Hoang, N., Gonzalez, M., Quy le, T., & Zhang, W. 2024. Fairaid: Navigating fairness, bias, and ethics in educational AI applications. arXiv preprint arXiv:2407.18745.

Cohausz, L., Kappenberger, J., & Stuckenschmidt, H. 2024. What fairness metrics can really tell you: A case study in the educational domain, in: *Proceedings of the 14th Learning Analytics and Knowledge Conference*, pp. 792–799.

Culp, J., & Drerup, J. 2024. Equality of educational opportunity, in: *Handbook of Equality of Opportunity*. Springer, pp. 521–545.

Deho, O. B., Zhan, C., Li, J., Liu, J., Liu, L., & Duy Le, T. 2022. How do the existing fairness metrics and unfairness mitigation algorithms contribute to ethical learning analytics? *British Journal of Educational Technology*

Deshpande, I., Zhang, Z., & Schwing, A. G. 2018. Generative modeling using the sliced wasserstein distance, in: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3483–3491.

Dikli, S., & Bleyle, S. 2014. Automated essay scoring feedback for second language writers: How does it compare to instructor feedback? *Assessing writing* 22, 1–17.

Dominguez-Catena, I., Paternain, D., & Galar, M. 2024. Metrics for dataset demographic bias: A case study on facial expression recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*

Efron, B. 1987. Better bootstrap confidence intervals. *Journal of the American statistical Association* 82, 171–185.

Friedman, J. N., Sacerdote, B., Staiger, D. O., & Tine, M. 2025. *Standardized test scores and academic performance at ivy-plus colleges*. Technical Report. National Bureau of Economic Research.

Gardner, J., Brooks, C., & Baker, R. 2019. Evaluating the fairness of predictive student models through slicing analysis, in: *Proceedings of the 9th international conference on learning analytics & knowledge*, pp. 225–234.

Gibbons, J. D., & Chakraborti, S. 2014. *Nonparametric statistical inference: revised and expanded*. CRC press.

Gibbs, A. L., & Su, F. E. 2002. On choosing and bounding probability metrics. *International statistical review* 70, 419–435.

Glaser, R., Chudowsky, N., & Pellegrino, J. W. 2001. *Knowing what students know: The science and design of educational assessment*. National Academies Press.

Gulson, K., Thompson, G., Swist, T., Kitto, K., Rutkowski, L., Rutkowski, D., Hogan, A., Zhang, V., & Knight, S. 2022. Automated essay scoring in australian schools: key issues and recommendations. white paper.

Gursory, F., & Kakadiaris, I. A. 2022. Error parity fairness: Testing for group fairness in regression tasks. arXiv preprint arXiv:2208.08279.

Han, X., Baldwin, T., & Cohn, T. 2022. Balancing out bias: Achieving fairness through balanced training, in: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, Abu Dhabi, United Arab Emirates. pp. 11335–11350.

Hardt, M., Price, E., & Srebro, N. 2016. Equality of opportunity in supervised learning. *NeurIPS* 29.

Incognito, O., & Pinto, G. 2023. Longitudinal effects of family and school context on the development on emergent literacy skills in preschoolers. *Current Psychology* 42, 9819–9829.

Jiang, H., & Nachum, O. 2020. Identifying and correcting label bias in machine learning, in: *International Conference on Artificial Intelligence and Statistics*, PMLR. pp. 702–712.

Kamiran, F., & Calders, T. 2012. Data preprocessing techniques for classification without discrimination. *Knowledge and information systems* 33, 1–33.

- Kamiran, F., Calders, T., & Pechenizkiy, M. 2010. Discrimination aware decision tree learning, in: *2010 IEEE international conference on data mining*, IEEE. pp. 869–874.
- Kizilcec, R. F., & Lee, H. 2022. Algorithmic fairness in education, in: *The Ethics of AIED*. Routledge, pp. 174–202.
- Kuncel, N. R., & Hezlett, S. A. 2007. Standardized tests predict graduate students' success. *Science* 315, 1080–1081.
- Lahoti, P., Beutel, A., Chen, J., Lee, K., Prost, F., Thain, N., Wang, X., & Chi, E. 2020. Fairness without demographics through adversarially reweighted learning. *Advances in neural information processing systems* 33, 728–740.
- Lallé, S., Bouchet, F., Verger, M., & Luengo, V. 2024. Fairness of mooc completion predictions across demographics and contextual variables, in: *Artificial Intelligence in Education: 25th International Conference, AIED 2024, Recife, Brazil, July 8–12, 2024, Proceedings, Part I*, Springer-Verlag, Berlin, Heidelberg. p. 379–393.
- Lee, H., Belitz, C., Nasir, N., & Bosch, N. 2025. Xai reveals the causes of attention deficit hyperactivity disorder (adhd) bias in student performance prediction, in: *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, pp. 418–428.
- Li, L., Sha, L., Li, Y., Raković, M., Rong, J., Joksimovic, S., Selwyn, N., Gašević, D., & Chen, G. 2023. Moral machines or tyranny of the majority? a systematic review on predictive bias in education, in: *LAK23: 13th International Learning Analytics and Knowledge Conference*, pp. 499–508.
- Li, L., Srivastava, N., Rong, J., Guan, Q., Gašević, D., & Chen, G. 2025. When and how biases seep in: Enhancing debiasing approaches for fair educational predictive analytics. *British Journal of Educational Technology*
- Li, L., Srivastava, N., Rong, J., Pianta, G., Varanasi, R., Gašević, D., & Chen, G. 2024. Unveiling goods and bads: A critical analysis of machine learning predictions of standardized test performance in early childhood education, in: *Proceedings of the 14th Learning Analytics and Knowledge Conference*, Association for Computing Machinery, New York, NY, USA. p. 608–619. <https://doi.org/10.1145/3636555.3636920>.
- Litman, D., Zhang, H., Correnti, R., Matsumura, L. C., & Wang, E. 2021. A fairness evaluation of automated methods for scoring text evidence usage in writing, in: *AIED*, Springer. pp. 255–267.
- Liu, H., Li, Y., Xing, T., Wang, P., Dalal, V., Li, L., He, J., & Wang, H. 2025a. Dataset distillation via the wasserstein metric, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1205–1215.
- Liu, Q., Deho, O., Vadiée, F., Khalil, M., Joksimovic, S., & Siemens, G. 2025b. Can synthetic data be fair and private? a comparative study of synthetic data generation and fairness algorithms, in: *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, pp. 591–600.
- Loukina, A., Madnani, N., & Zechner, K. 2019. The many dimensions of algorithmic fairness in educational applications, in: *Proceedings of the Fourteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pp. 1–10.
- McGaw, B., Loudon, W., & Wyatt-Smith, C. 2020. *NAPLAN review final report*. [Commissioned by]: State of New South Wales (Department of Education), State
- Mining, W. I. D. 2006. Data mining: Concepts and techniques. *Morgan Kaufmann* 10, 4.
- Mohamed, M. M., & Schuller, B. W. 2022. Normalise for fairness: A simple normalisation technique for fairness in regression machine learning problems. arXiv preprint arXiv:2202.00993.
- Montgomery, D. C., Peck, E. A., & Vining, G. G. 2021. *Introduction to linear regression analysis*. John Wiley & Sons.
- Murtagh, F. 1991. Multilayer perceptrons for classification and regression. *Neurocomputing* 2, 183–197.
- Oliveira, H., Ferreira Mello, R., Barreiros Rosa, B. A., Rakovic, M., Miranda, P., Cordeiro, T., Isotani, S., Bittencourt, I., & Gasevic, D. 2023. Towards explainable prediction of essay cohesion in portuguese and english, in: *LAK23*, pp. 509–519.
- Panaretos, V. M., & Zemel, Y. 2019. Statistical aspects of wasserstein distances. *Annual review of statistics and its application* 6, 405–431.
- Rassouli, B., & Gündüz, D. 2019. Optimal utility-privacy trade-off with total variation distance as a privacy measure. *IEEE Transactions on Information Forensics and Security* 15, 594–603.
- Ritter, S., Joshi, A., Fancsali, S., & Nixon, T. 2013. Predicting standardized test scores from cognitive tutor interactions, in: *Educational Data Mining 2013*.
- Sha, L., Raković, M., Das, A., Gašević, D., & Chen, G. 2022. Leveraging class balancing techniques to alleviate algorithmic bias for predictive tasks in education. *IEEE Transactions on Learning Technologies*
- Shen, J., Qu, Y., Zhang, W., & Yu, Y. 2018. Wasserstein distance guided representation learning for domain adaptation, in: *Proceedings of the AAAI conference on artificial intelligence*.
- R Core Team and others 2021. R: A language and environment for statistical computing. *R foundation for statistical computing*, Vienna, Austria
- Thomson, S. 2018. Achievement at school and socioeconomic background—an educational perspective. *npj Science of Learning* 3, 5.
- Verger, M., Lallé, S., Bouchet, F., & Luengo, V. 2023. Is your model "madd"? a novel metric to evaluate algorithmic fairness for predictive student models. arXiv preprint arXiv:2305.15342.
- Wickham, H. 2011. ggplot2. *Wiley interdisciplinary reviews: computational statistics* 3, 180–185.
- William, D. 2010. Standardized testing and school accountability. *Educational Psychologist* 45, 107–122.
- Yang, K., Raković, M., Li, Y., Guan, Q., Gašević, D., & Chen, G. 2024. Unveiling the tapestry of automated essay scoring: A comprehensive investigation of accuracy, fairness, and generalizability, in: *Proceedings of the aaai conference on artificial intelligence*, pp. 22466–22474.
- Yao, C., Cortez, C., & Yu, R. 2025. Towards fair and privacy-aware transfer learning for educational predictive modeling: A case study on retention prediction in community colleges, in: *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, pp. 738–749.
- Yu, Z., Chakraborty, J., & Menzies, T. 2021. Fairbalance: Improving machine learning fairness on multiplesensitive attributes with data balancing. arXiv preprint arXiv:2107.08310.